AUTOMATED TESTING TOOLS (SELENIUM)

Installing and Creating Your First Application in ECLIPSE:

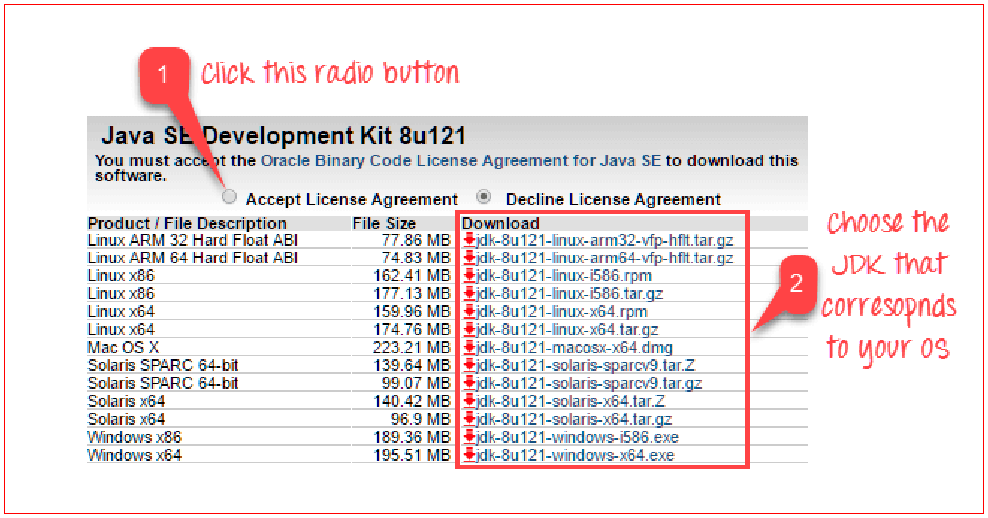
Selenium installation is a 3 step process:

1. Install Java SDK
2. Install Eclipse
3. Install Selenium Web driver Files

Step 1 – Install Java on your computer

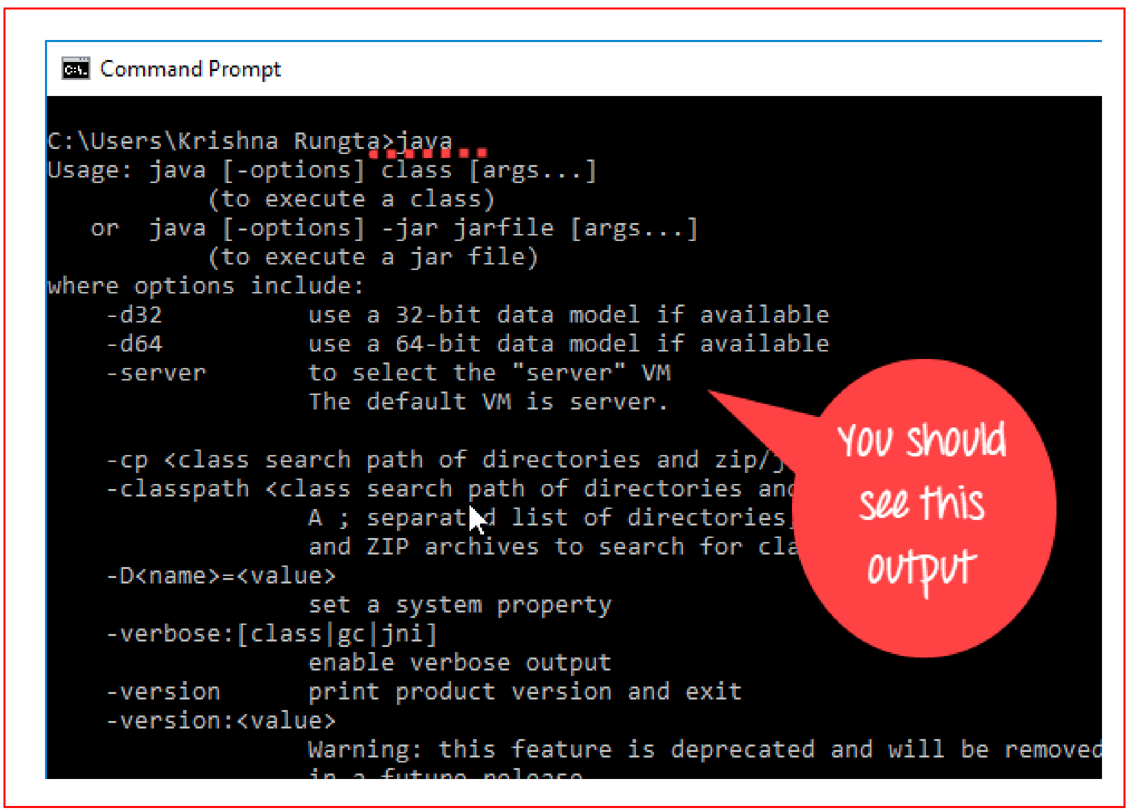
Download and install the **Java Software Development Kit (JDK)**

Next –



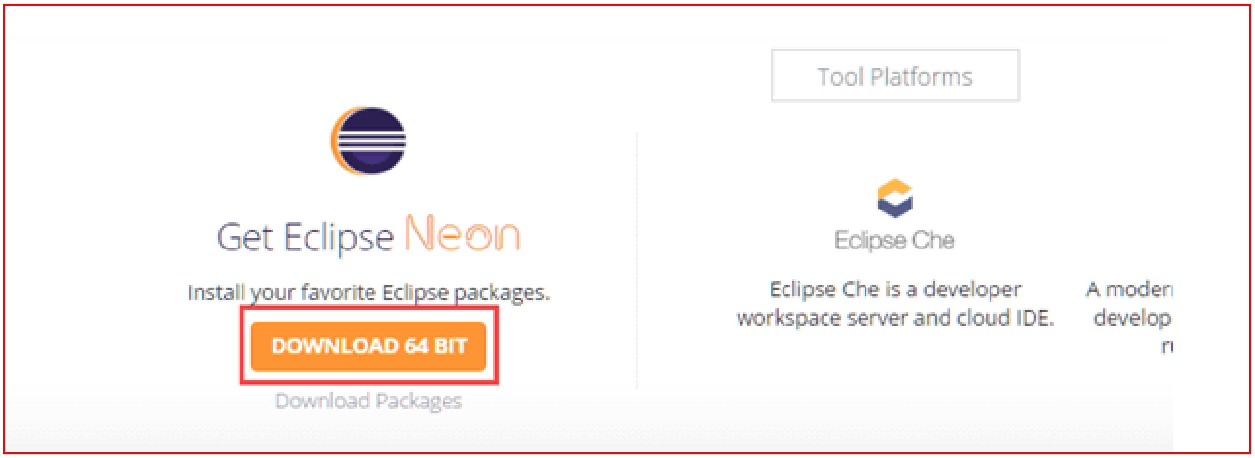
This JDK version comes bundled with Java Runtime Environment (JRE), so you do not need to download and install the JRE separately.

Once installation is complete, open command prompt and type “java”. If you see the following screen you are good to move to the next step

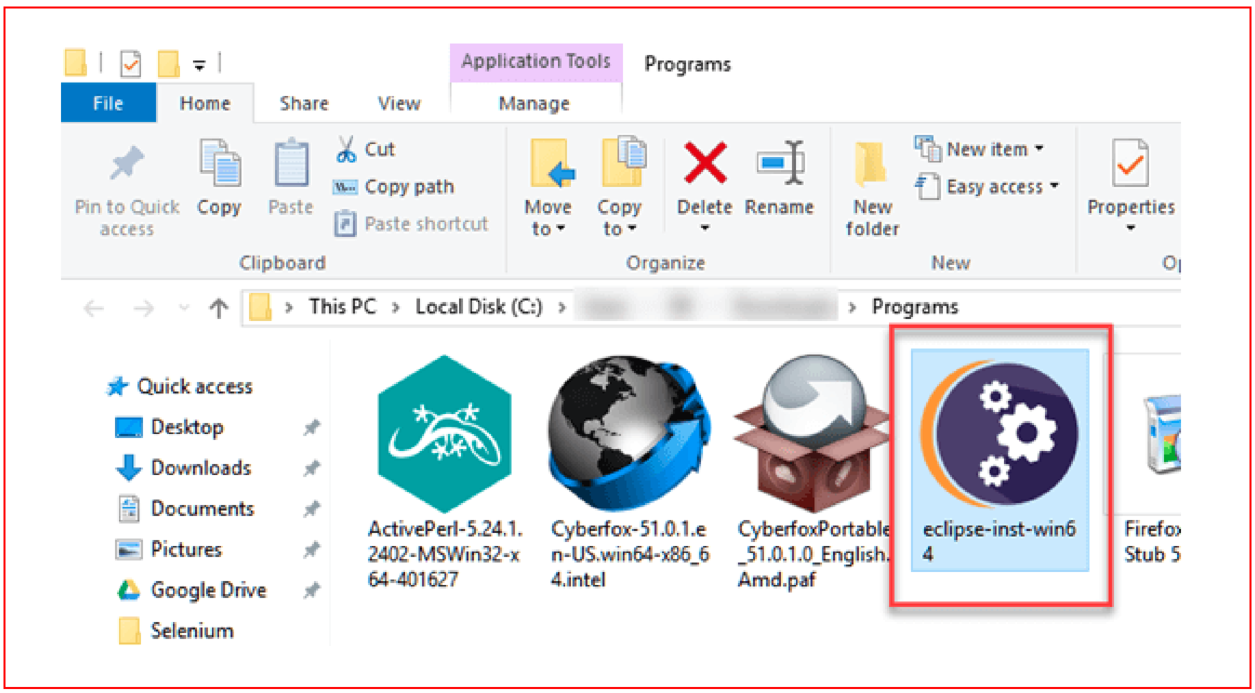


Step 2 – Install Eclipse IDE

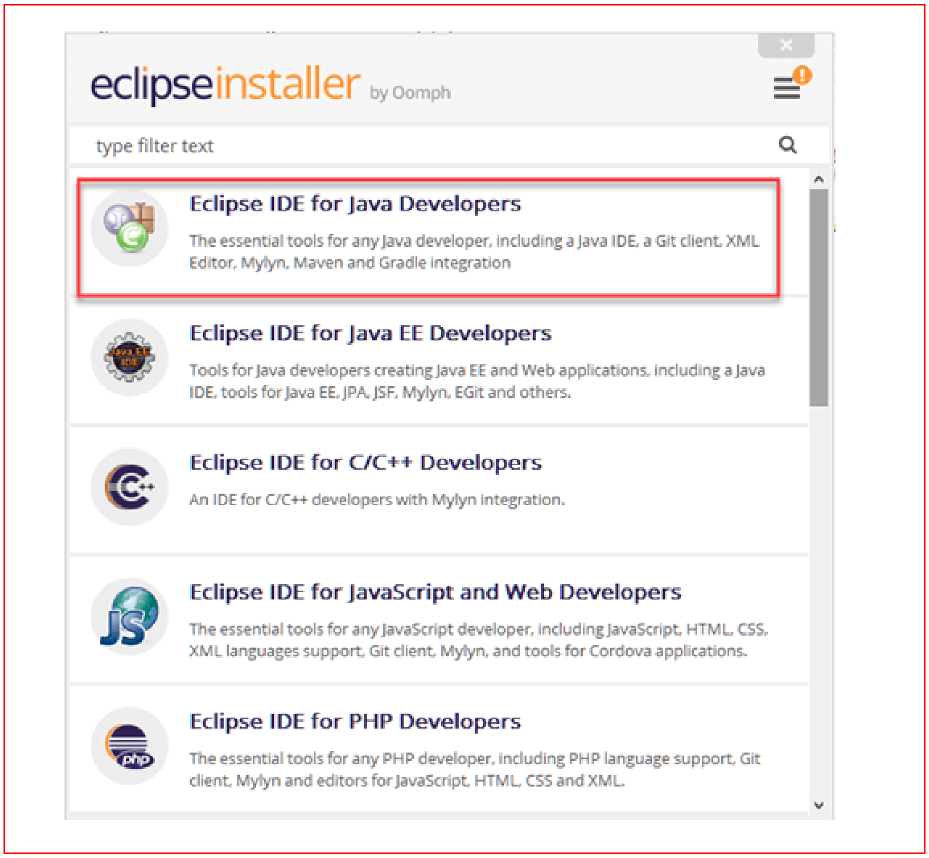
Download latest version of **"Eclipse IDE for Java Developers"** here. Be sure to choose correctly between Windows 32 Bit and 64 Bit versions.



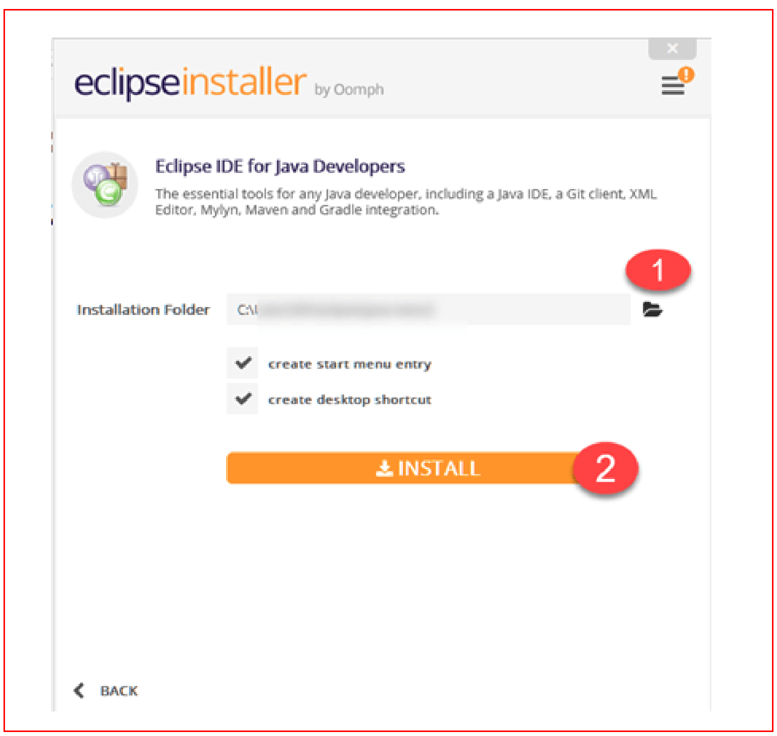
You should be able to download an exe file named "eclipse-inst-win64" for Setup.



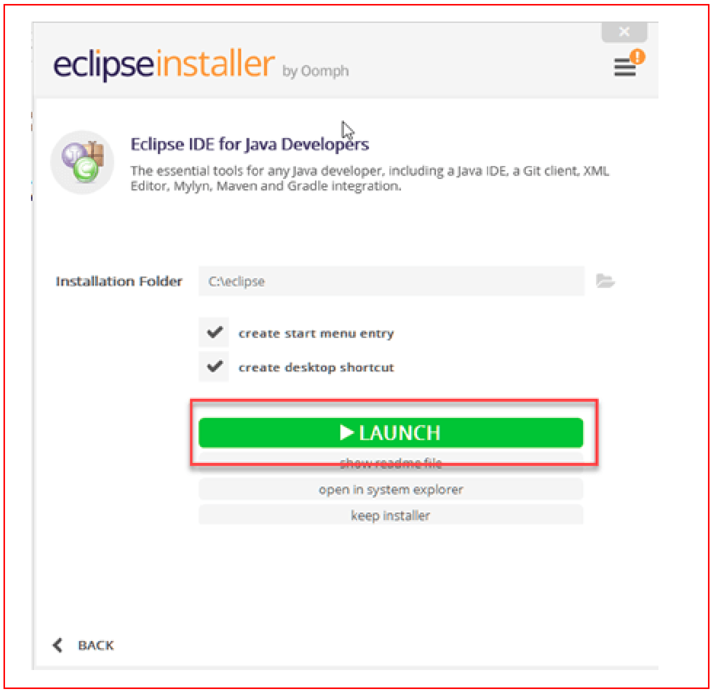
Double-click on file to Install the Eclipse. A new window will open. Click Eclipse IDE for Java Developers.



After that, a new window will open which click button marked 1 and change path to “C:\eclipse”. Post that Click on Install button marked 2



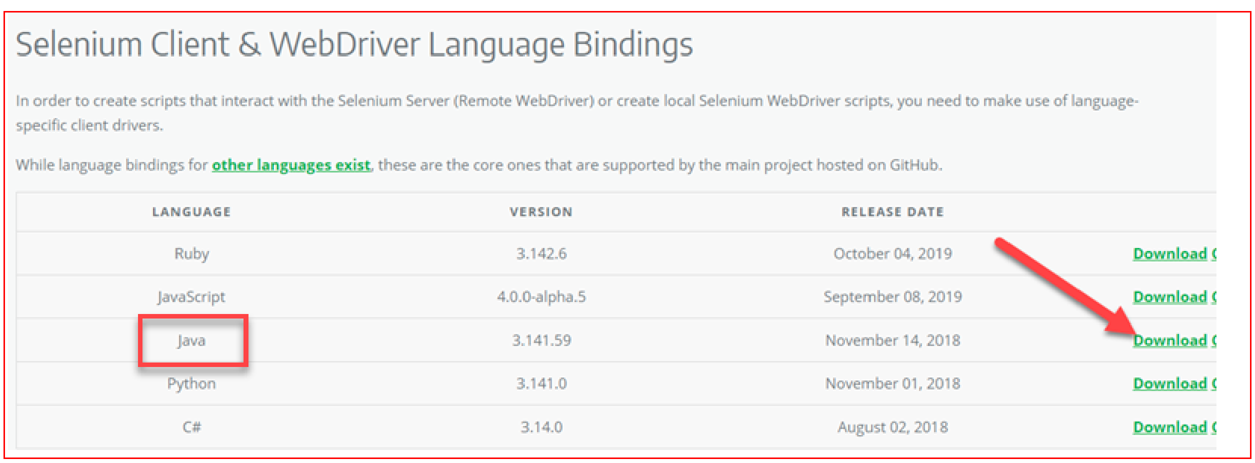
After successful completion of the installation procedure, a window will appear. On that window click on Launch



This will start eclipse neon IDE for you.

Step 3 – Download the Selenium Java Client Driver

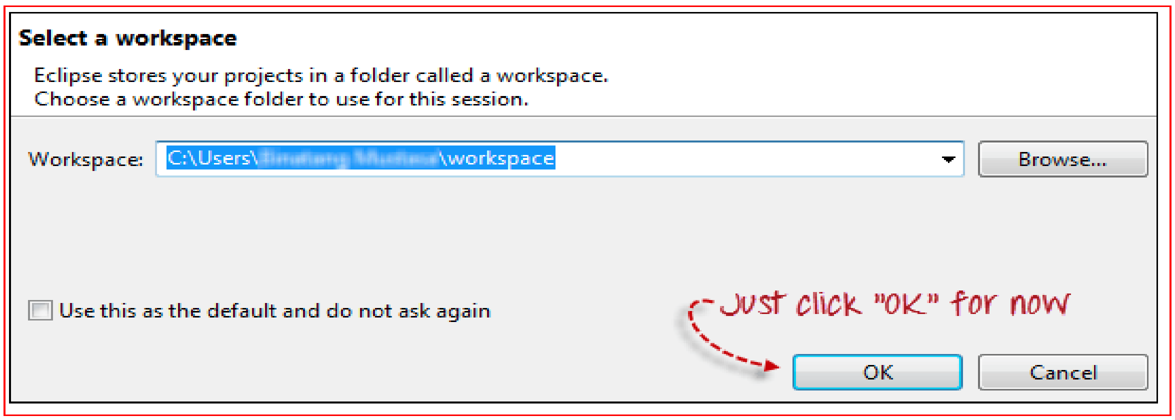
You can download **Selenium Webdriver for Java Client Driver** here. You will find client drivers for other languages there, but only choose the one for Java.



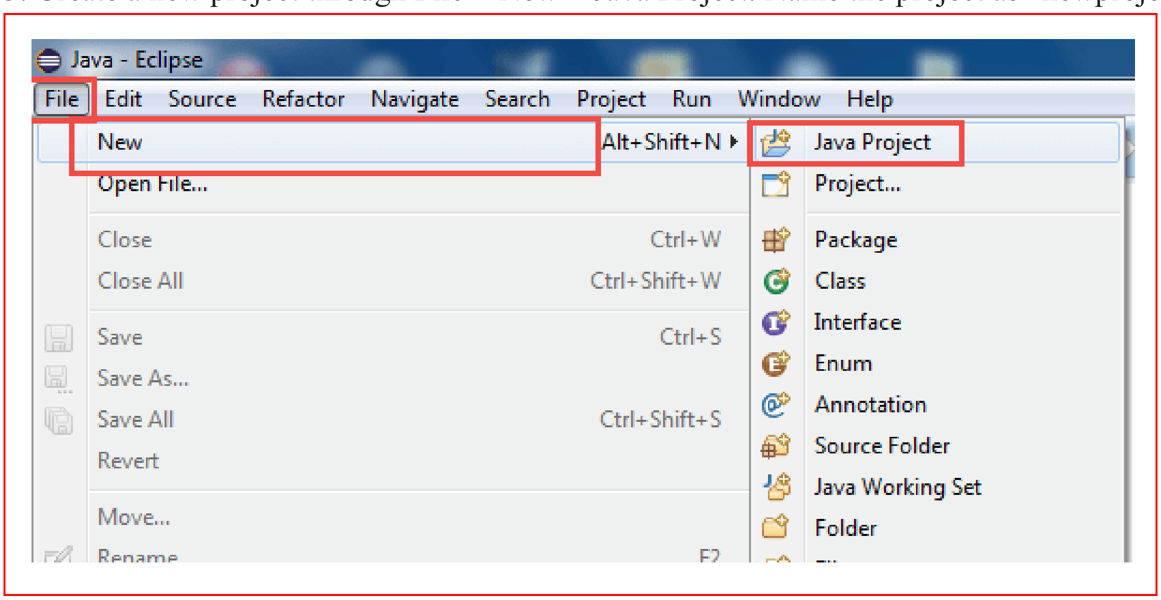
This download comes as a ZIP file named “selenium-3.14.0.zip”. For simplicity of Selenium installation on Windows 10, extract the contents of this ZIP file on your C drive so that you would have the directory "C:\selenium-3.14.0\". This directory contains all the JAR files that we would later import on Eclipse for Selenium setup.

Step 4 – Configure Eclipse IDE with WebDriver

1. Launch the "eclipse.exe" file inside the "eclipse" folder that we extracted in step 2. If you followed step 2 correctly, the executable should be located on C:\eclipse\eclipse.exe.
2. When asked to select for a workspace, just accept the default location.



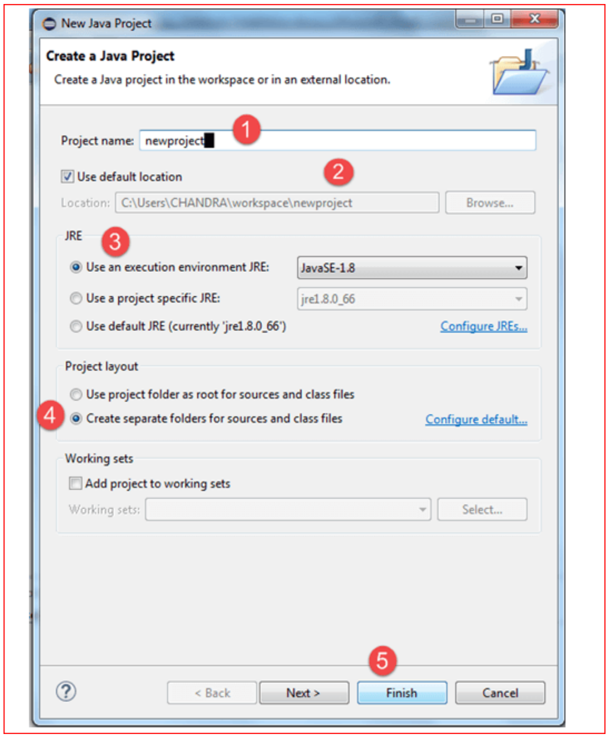
3. Create a new project through File > New > Java Project. Name the project as "newproject".



A new pop-up window will open enter details as follow

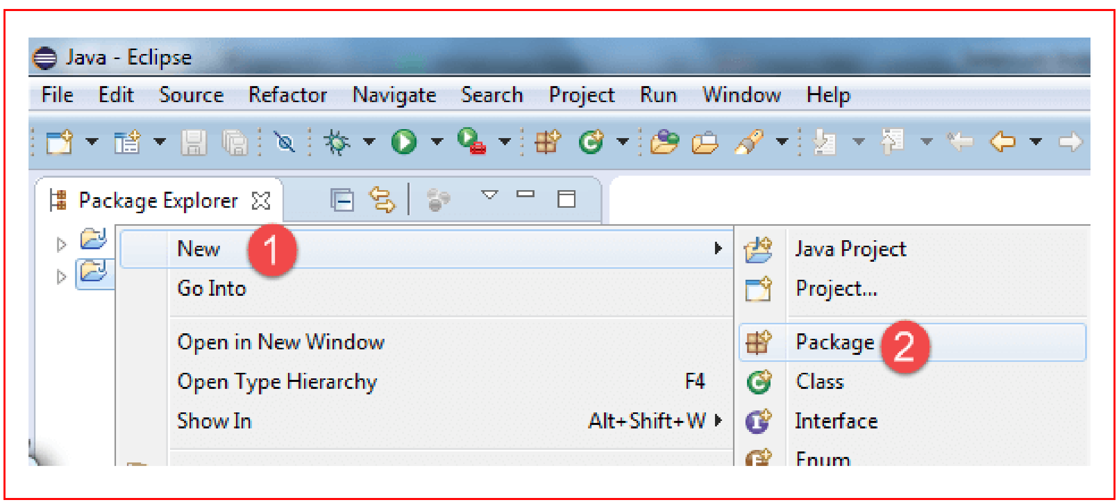
1. Project Name
2. Location to save project

3. Select an execution JRE
4. Select layout project option
5. Click on Finish button



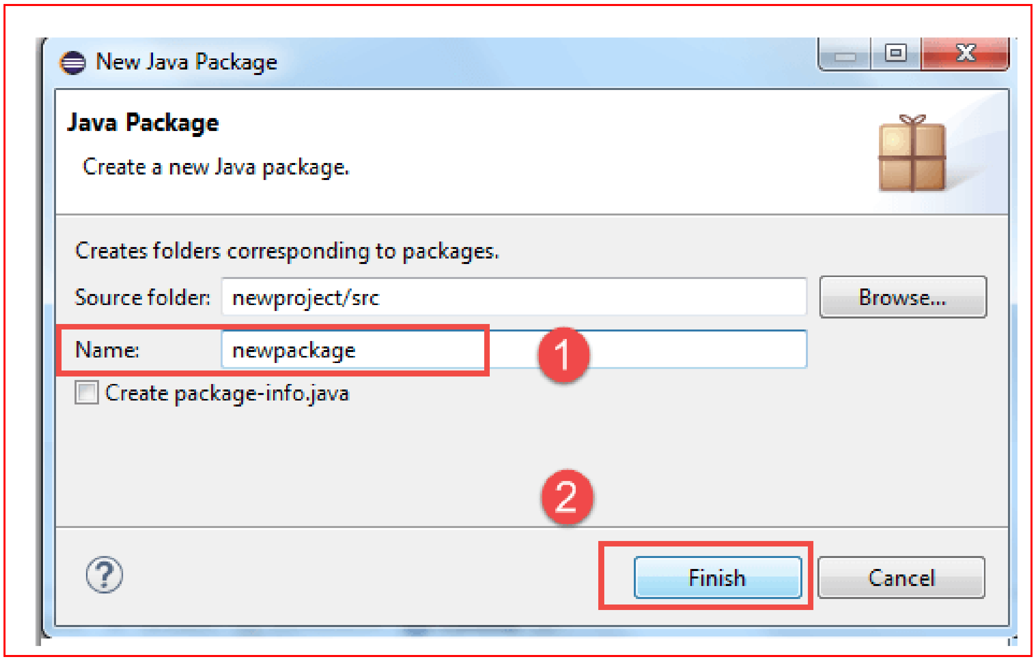
4. In this step,

1. Right-click on the newly created project and
2. Select New > Package, and name that package as “newpackage”.

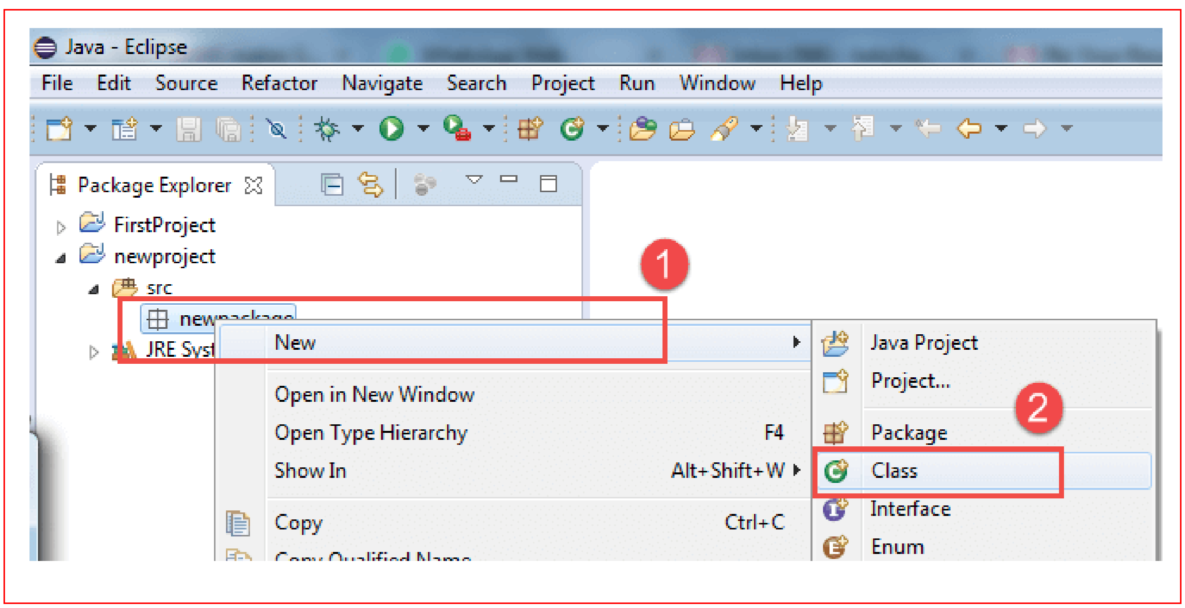


A pop-up window will open to name the package,

1. Enter the name of the package
2. Click on Finish button



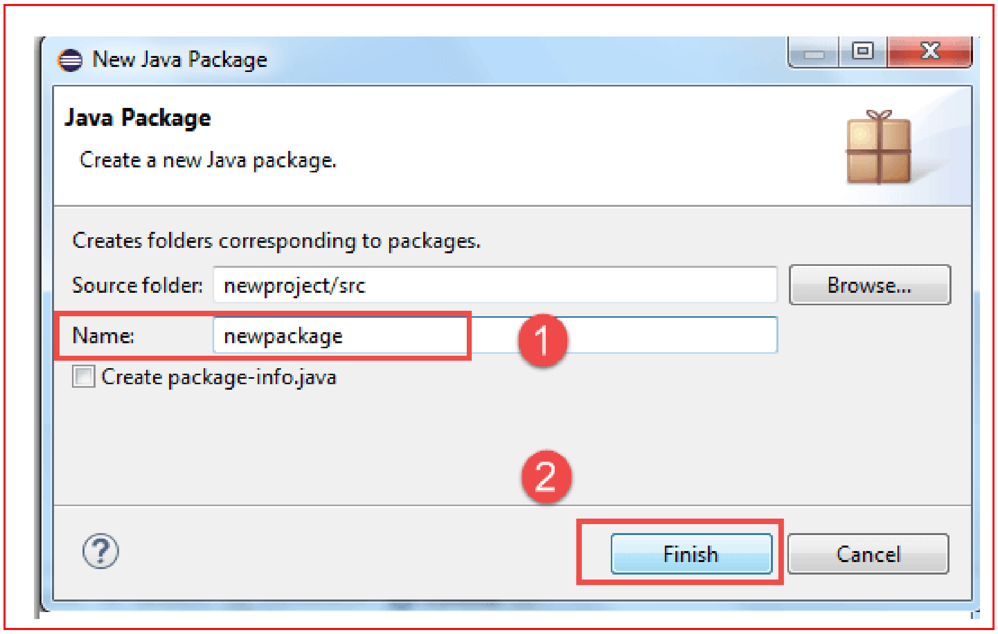
5. Create a new Java class under newpackage by right-clicking on it and then selecting- New > Class, and then name it as "MyClass". Your Eclipse IDE should look like the image below.



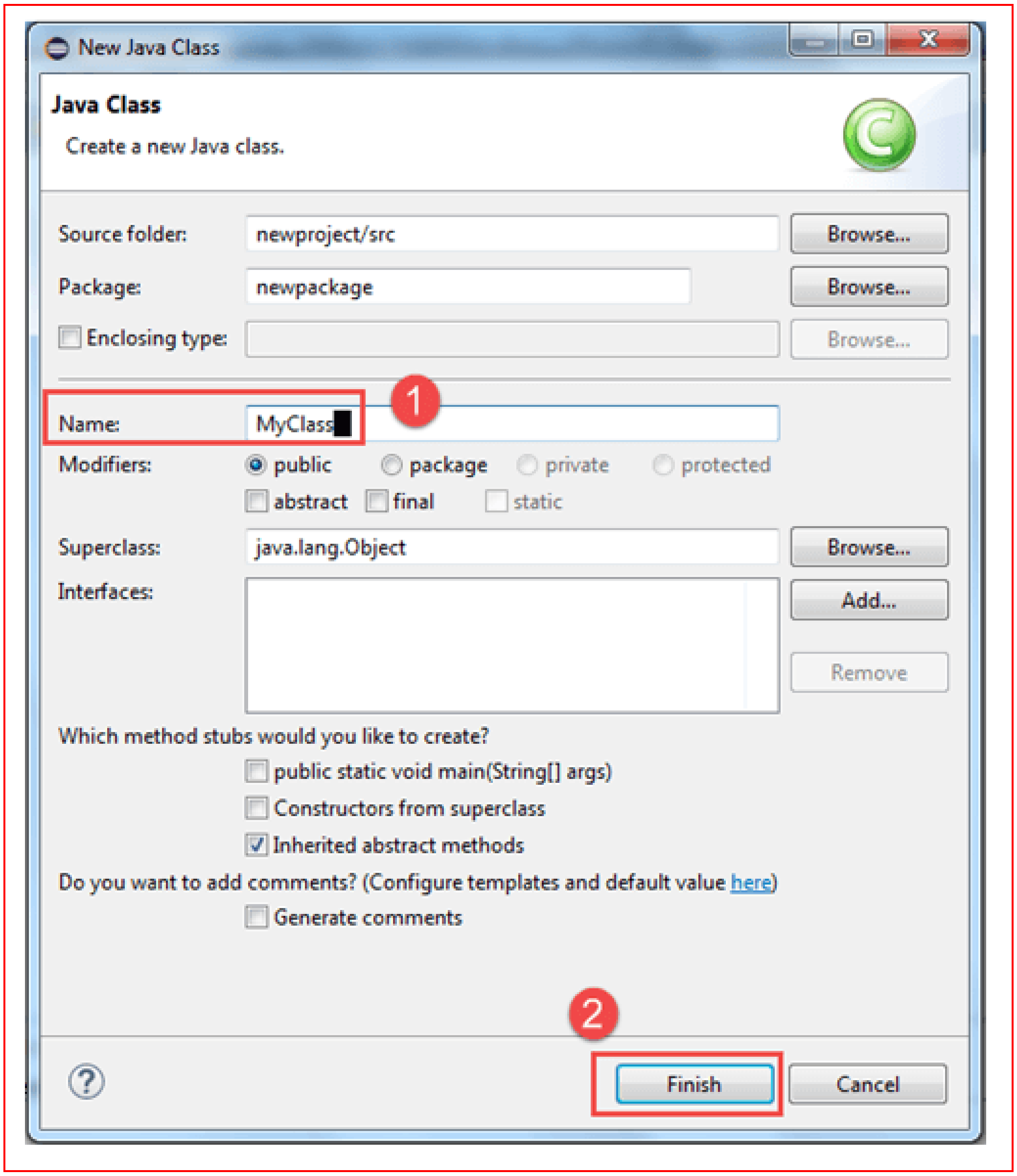
When you click on Class, a pop-up window will open, enter details as

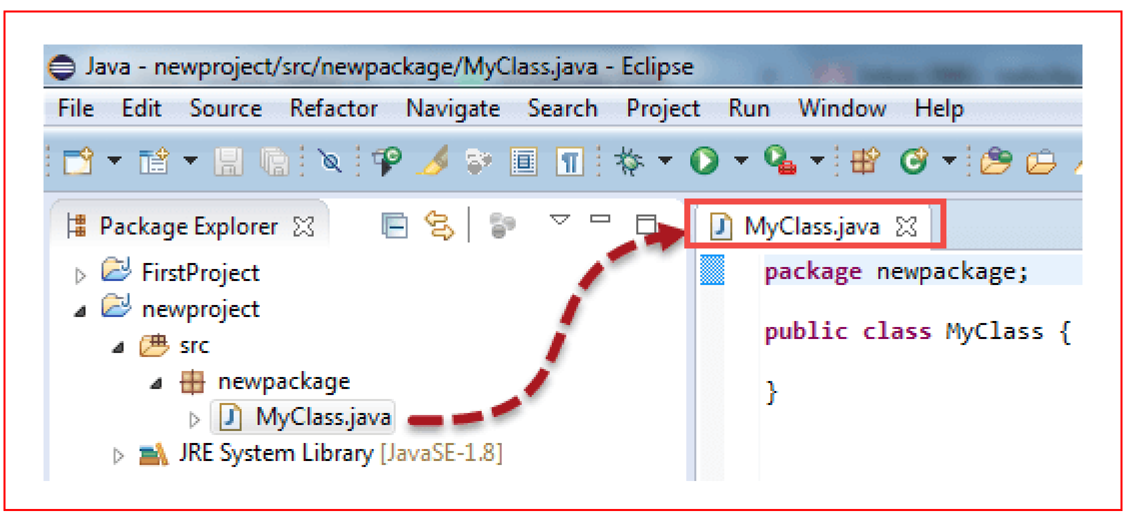
1. Name of the class

2. Click on Finish button



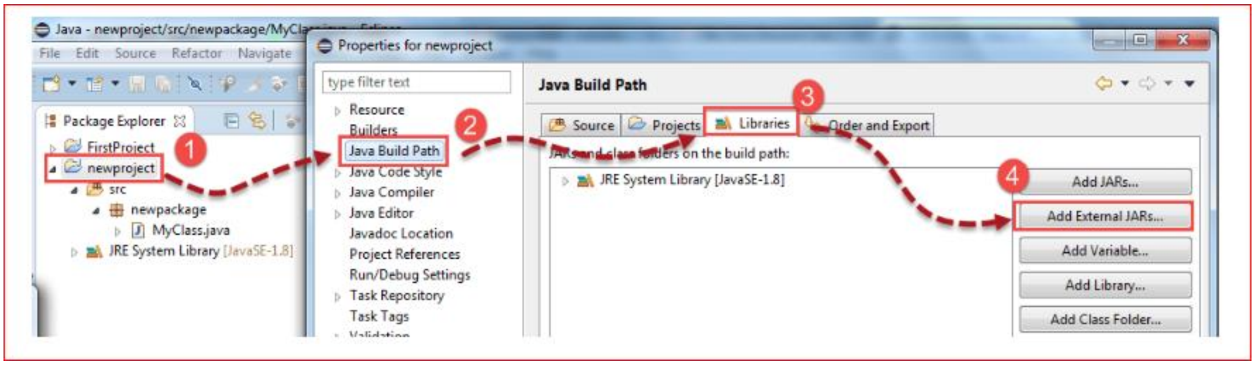
This is how it looks like after creating class.


New Java Class
Java Class
Create a new Java class.
Source folder: newproject/src
Browse...
Package: newpackage
Browse...
Enclosing type:
Browse...
1
Name: MyClass
Modifiers: public package private protected
abstract final static
Superclass: java.lang.Object
Browse...
Interfaces:
Add...
Remove
Which method stubs would you like to create?
public static void main(String[] args)
Constructors from superclass
Inherited abstract methods
Do you want to add comments? (Configure templates and default value here)
Generate comments
2
Finish
Cancel

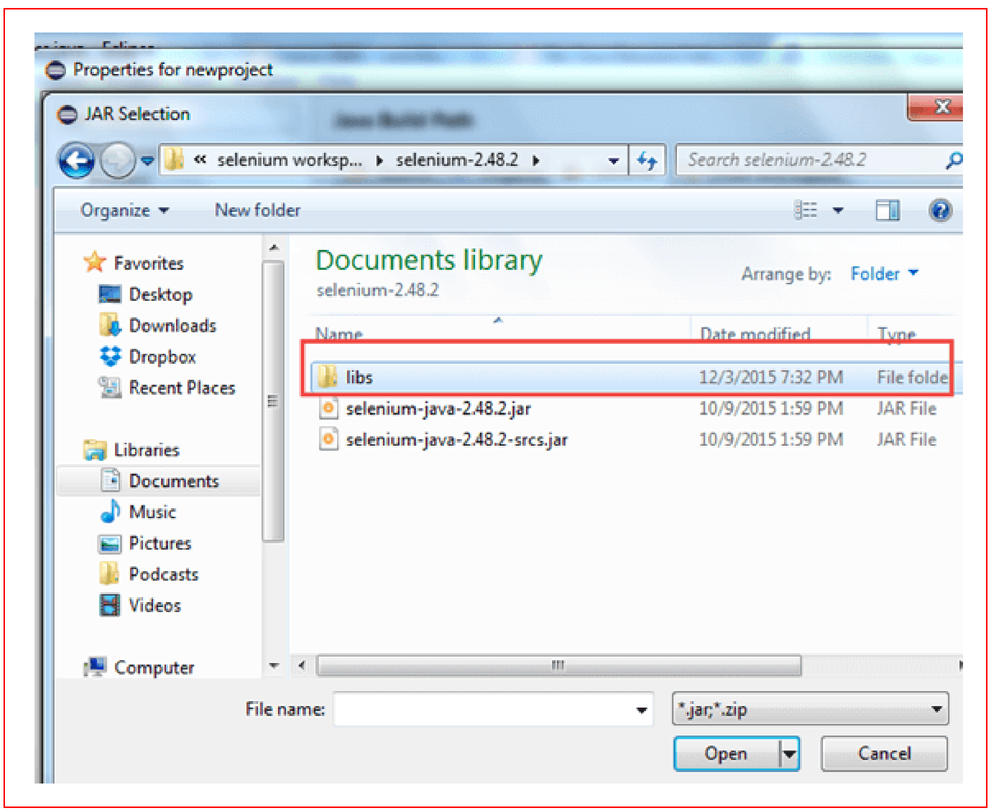

Java - newproject/src/newpackage/MyClass.java - Eclipse
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
FirstProject
newproject
src
newpackage
MyClass.java
JRE System Library [JavaSE-1.8]
MyClass.java
package newpackage;
public class MyClass {
}

Now selenium WebDriver's into Java Build Path
In this step,

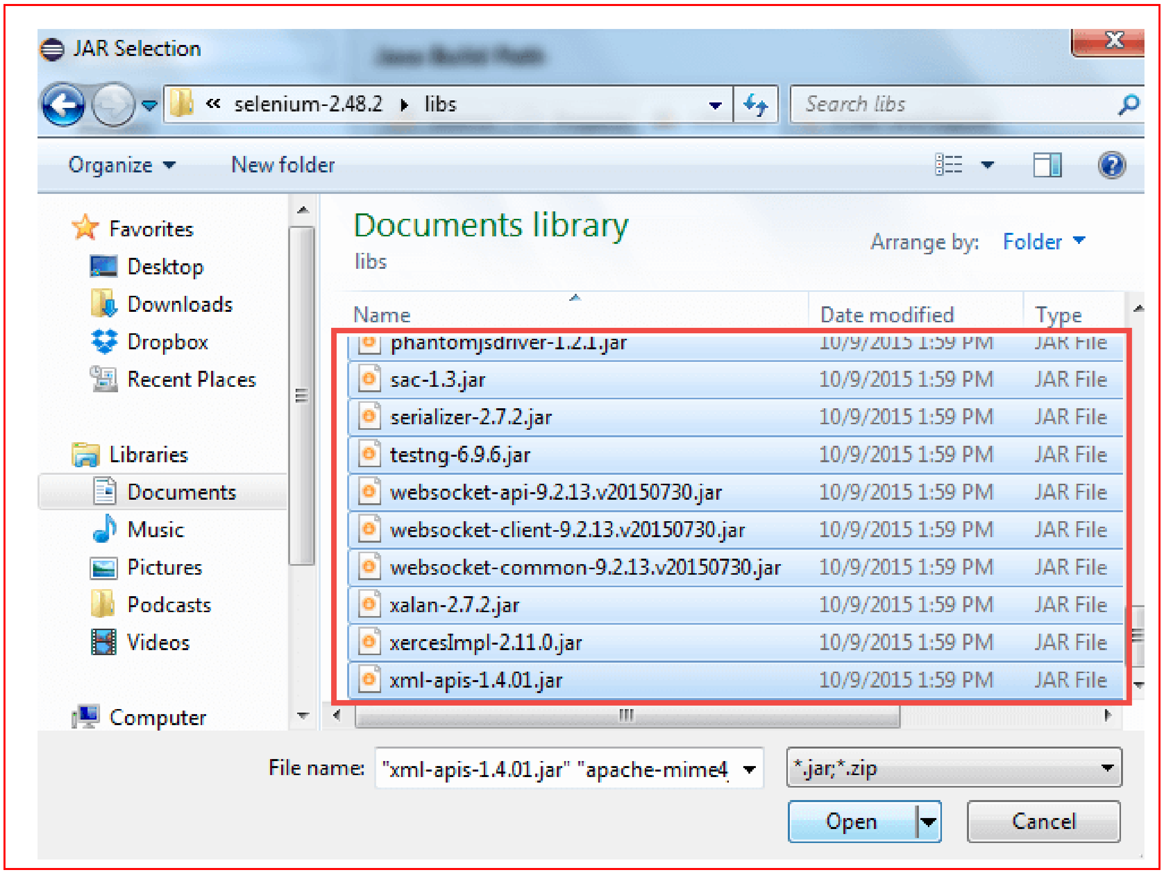
1. Right-click on "newproject" and select **Properties**.
2. On the Properties dialog, click on "Java Build Path".
3. Click on the **Libraries** tab, and then
4. Click on "Add External JARs.."



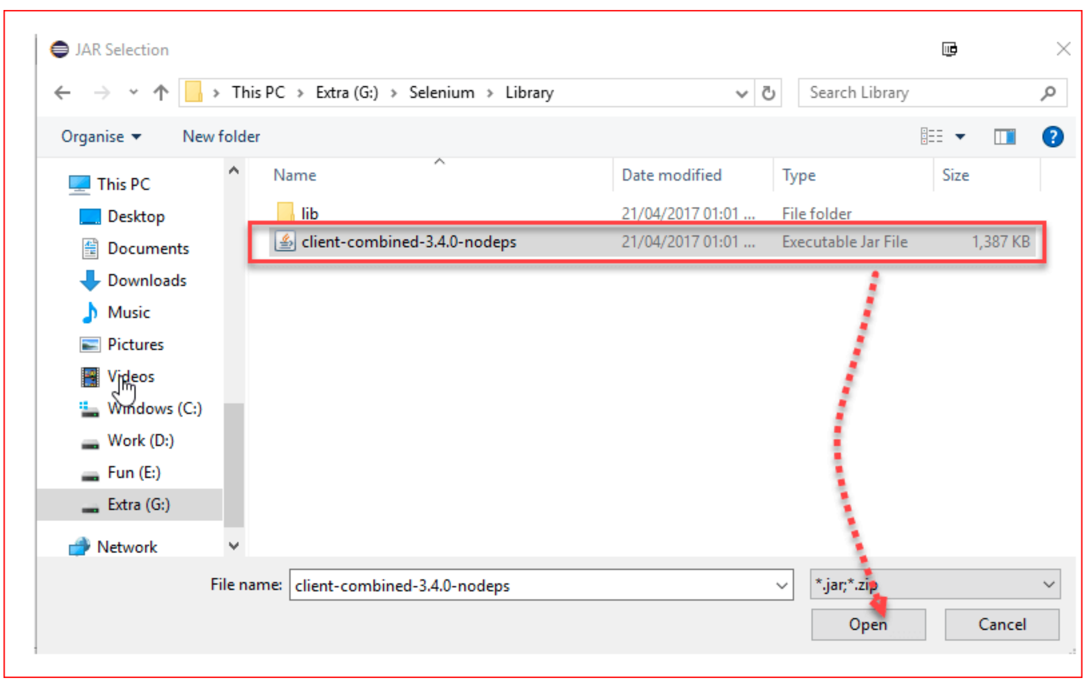
When you click on "Add External JARs.." It will open a pop-up window. Select the JAR files you want to add.



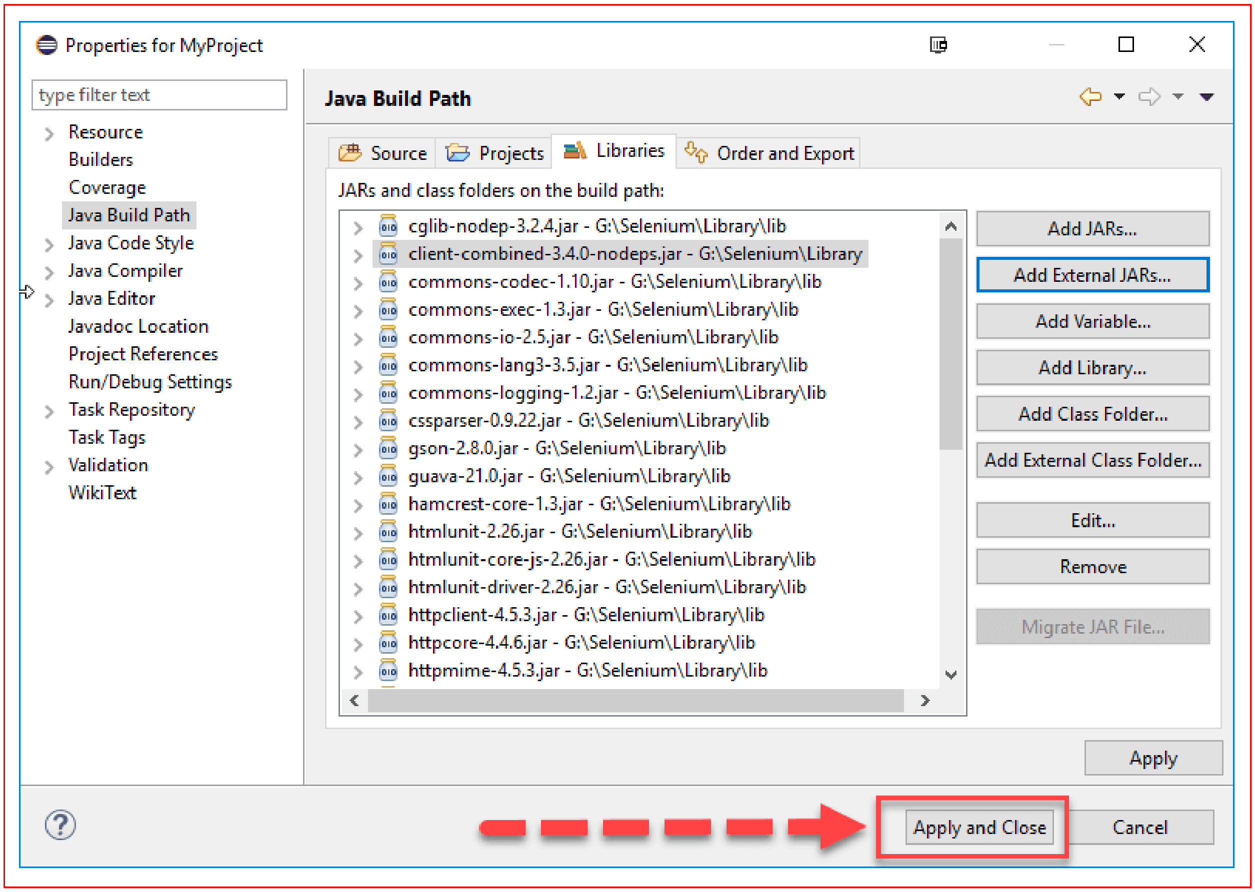
After selecting jar files, click on OK button.
Select all files inside the lib folder.



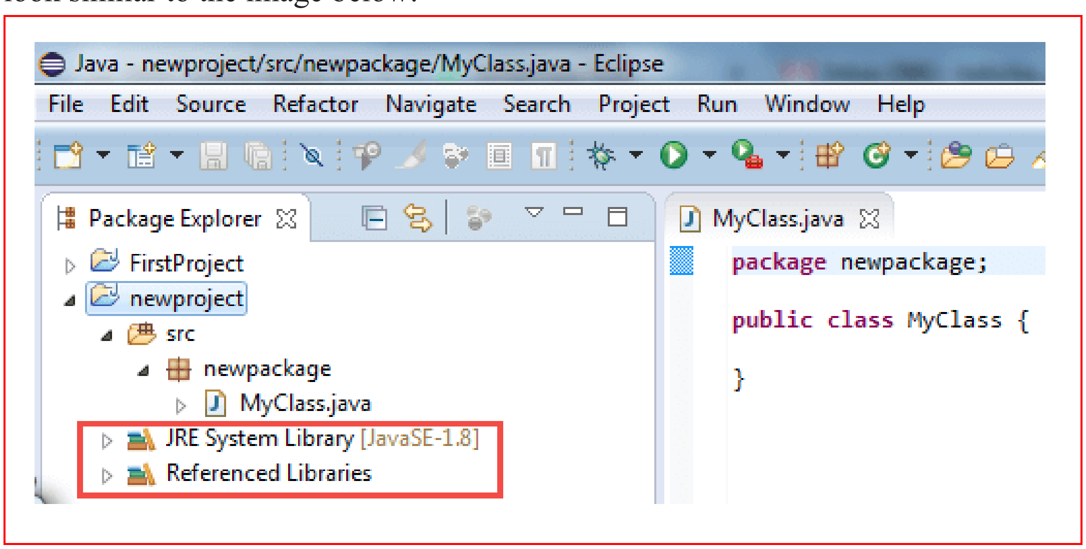
Select files outside lib folder



Once done, click "Apply and Close" button



6. Add all the JAR files inside and outside the "libs" folder. Your Properties dialog should now look similar to the image below.



7. Finally, click OK and we are done importing Selenium libraries into our project.

WEEK PROGRAMS

WEEK-1

AIM: Download and Install JAVA, Associate SWD Jars and Browser drivers.

ALGORITHM:

1. How to install JAVA
2. How to create and associate with SWD jars
3. How to select the browser drivers and make them to install.
4. How to convert browser drivers into .exe format.

SOURCE CODE:

Install Java on Windows 10

After downloading the installation file, proceed with installing Java on your Windows system.

Follow the steps below:

Step 1: Run the Downloaded File

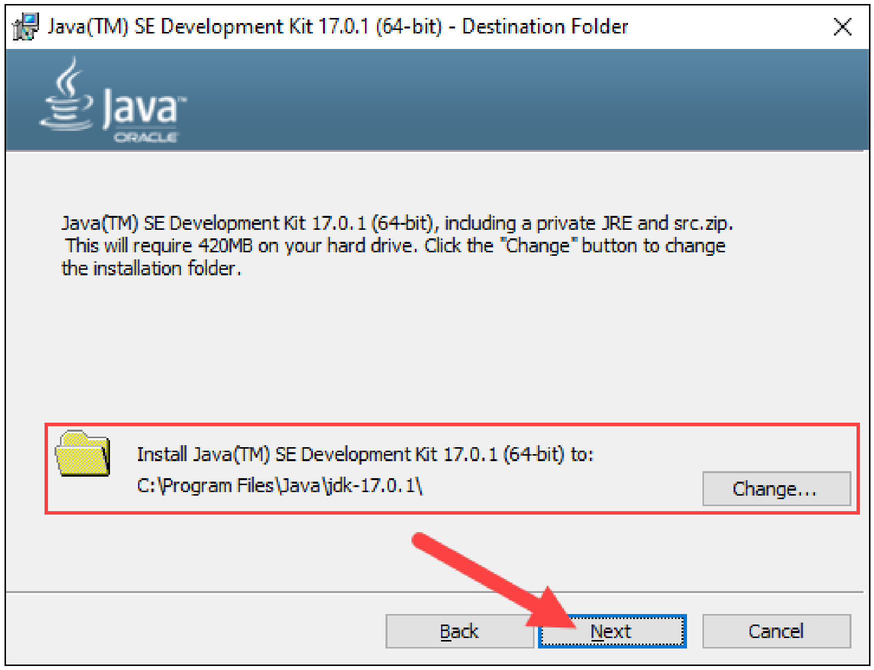
Double-click the **downloaded file** to start the installation.

Step 2: Configure the Installation Wizard

After running the installation file, the installation wizard welcome screen appears.

1. Click **Next** to proceed to the next step.

3. Choose the destination folder for the Java installation files or stick to the default path. Click **Next** to proceed.



5. Wait for the wizard to finish the installation process until the *Successfully Installed* message appears. Click **Close** to exit the wizard.

Associate SWD Jars : [selenium web driver]

Selenium Server (Grid)

The Selenium Server is needed in order to run Remote Selenium WebDriver (Grid).
Selenium Grid allows the execution of WebDriver scripts on remote machines (virtual or real) by routing commands sent by the client to remote browser instances. It aims to provide an easy way to run tests in parallel on multiple machines.

Selenium Grid allows us to run tests in parallel on multiple machines, and to manage different browser versions and browser configurations centrally (instead of in each individual test).

Selenium Grid is not a silver bullet. It solves a subset of common delegation and distribution problems, but will for example not manage your infrastructure, and might not suit your specific needs.

Purposes and main functionalities

- Central entry point for all tests
- Management and control of the nodes / environment where the browsers run
- Scaling
- Running tests in parallel
- Cross-platform testing
- Load balancing

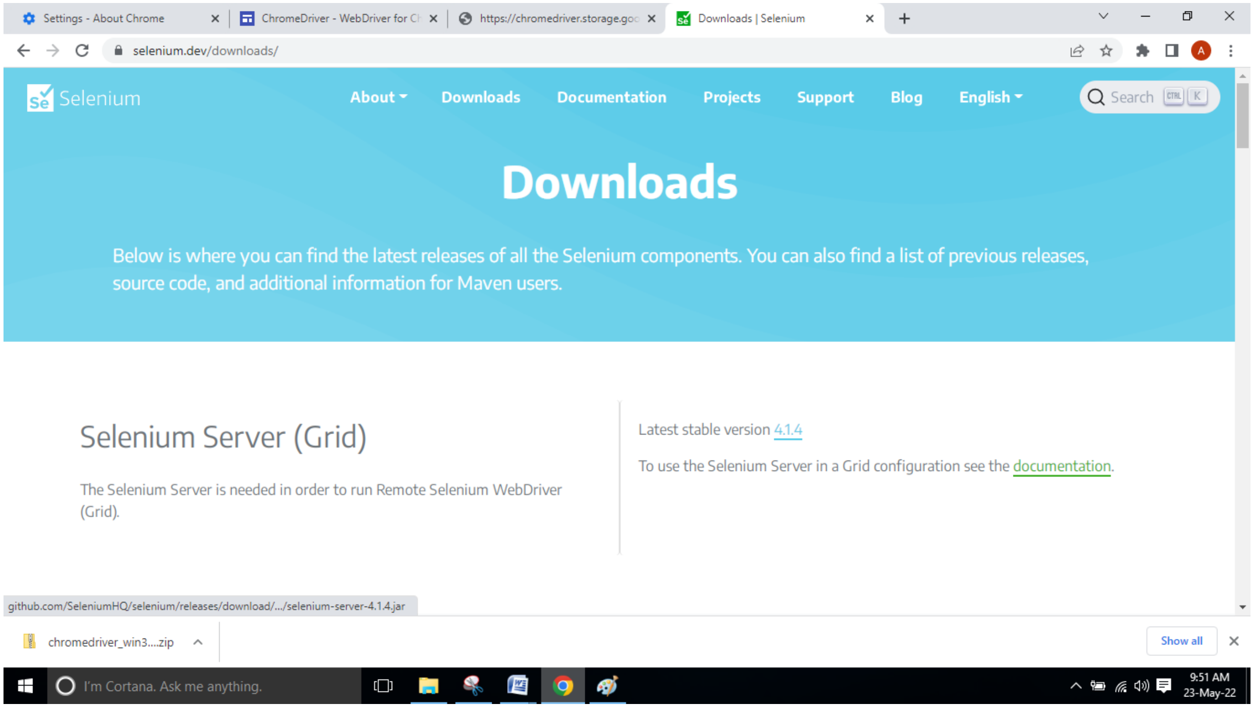
Selenium Grid 4

Grid 4 takes advantage of a number of new technologies in order to facilitate scaling up while allowing local execution.

Selenium Grid 4 is a fresh implementation and does not share the codebase the previous version had.

To get all the details of Grid 4 components, understand how it works, and how to set up you own, please browse thorough the following sections.

// Latest stable version 4.1.4



<u>Browser drivers :</u>

Install browser drivers

Setting up your system to allow a browser to be automated.

Through WebDriver, Selenium supports all major browsers on the market such as Chrome/Chromium, Firefox, Internet Explorer, Edge, Opera, and Safari. Where possible, WebDriver drives the browser using the browser's built-in support for automation.

Since all the driver implementations except for Internet Explorer are provided by the browser vendors themselves, they are not included in the standard Selenium distribution. This section explains the basic requirements for getting you started with the different browsers.

Three Ways to Use Drivers

1. Driver Management Software

Most machines automatically update the browser, but the driver does not. To make sure you get the correct driver for your browser, there are many third party libraries to assist you.

1. Import WebDriver Manager

```
import io.github.bonigarcia.wdm.WebDriverManager;
```

Copy

2. Calling `setup()` automatically puts the correct browser driver where the code will see it:

```
WebDriverManager.chromedriver().setup();
```
Copy

3. Just initialize the driver as you normally would:

```
ChromeDriver driver = new ChromeDriver()
```

2. The PATH Environment Variable

This option first requires manually downloading the driver.

This is a flexible option to change location of drivers without having to update your code, and will work on multiple machines without requiring that each machine put the drivers in the same place.

You can either place the drivers in a directory that is already listed in `PATH`, or you can place them in a directory and add it to `PATH`.
To see what directories are already on PATH, open a Command Prompt and execute:

```
echo %PATH%
```
Copy
If the location to your driver is not already in a directory listed, you can add a new directory to PATH:

```
setx PATH "%PATH%;C:\WebDriver\bin"
```
Copy
You can test if it has been added correctly by starting the driver:

```
chromedriver.exe
```
Copy
If your `PATH` is configured correctly above, you will see some output relating to the startup of the driver:

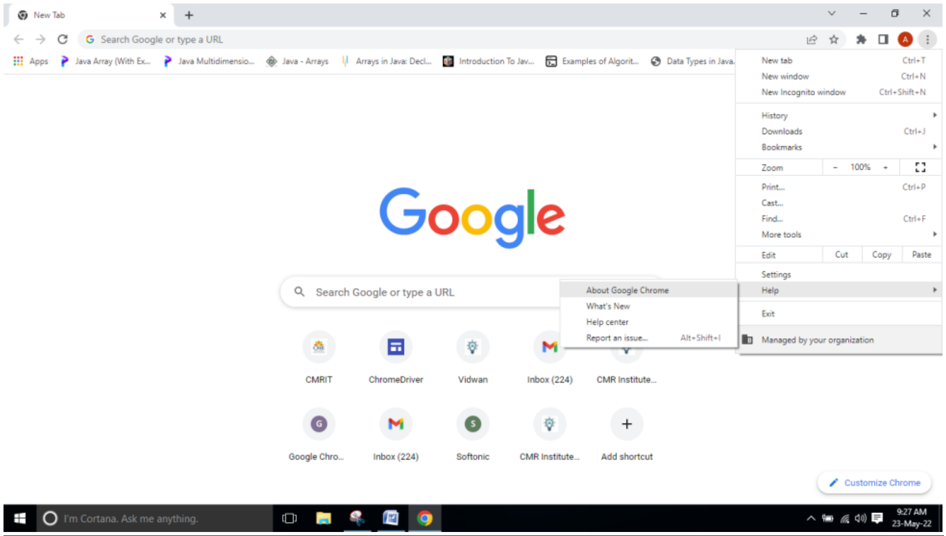
```
Starting ChromeDriver 95.0.4638.54 (d31a821ec901f68d0d34ccdbaea45b4c86ce543e-refs/branch-heads/4638@{#871})
on port 9515
Only local connections are allowed.
Please see https://chromedriver.chromium.org/security-considerations for suggestions on keeping ChromeDriver safe.
ChromeDriver was started successfully.
```

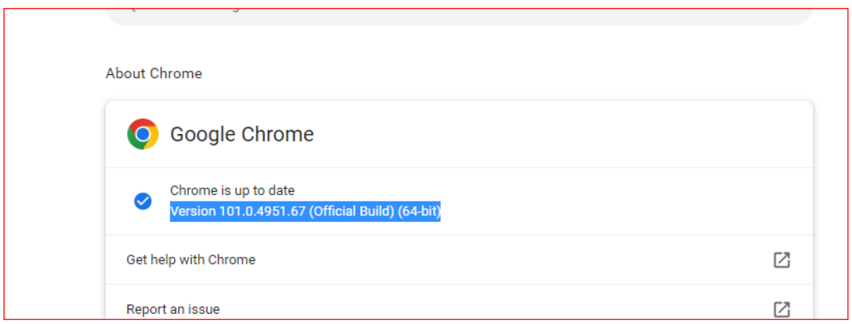
3. Hard Coded Location

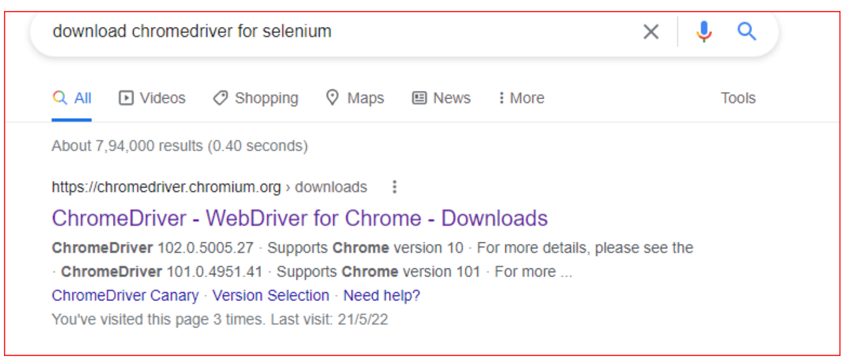
Similar to Option 2 above, you need to manually download the driver Specifying the location in the code itself has the advantage of not needing to figure out Environment Variables on your system, but has the drawback of making the code much less flexible.

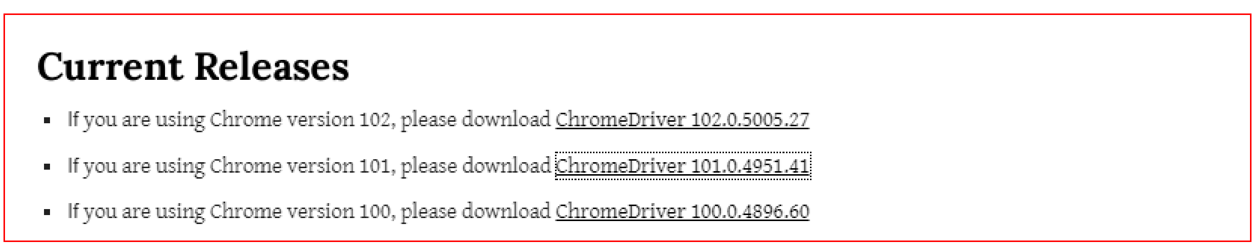
```
System.setProperty("webdriver.chrome.driver","/path/to/chromedriver");
ChromeDriver driver = new ChromeDriver();
```

STEPS TO INSTALL CHROME DRIVER :



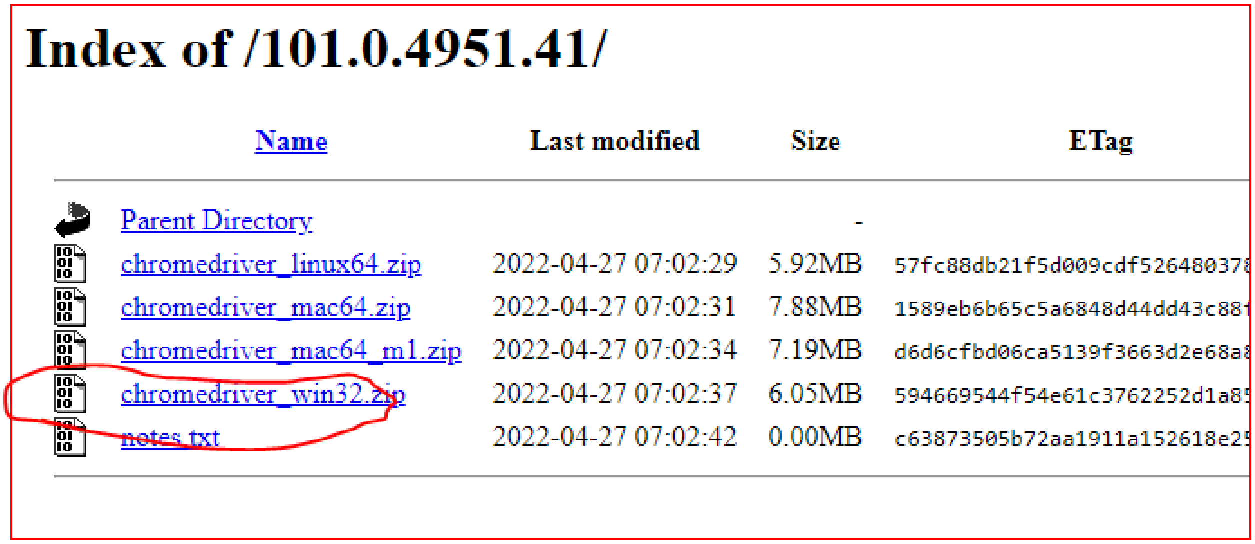


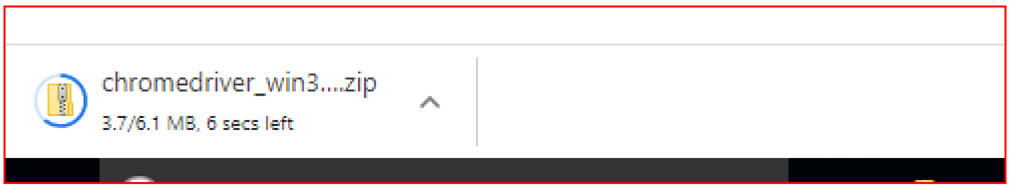


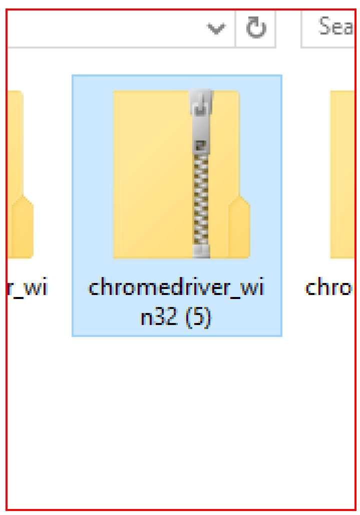


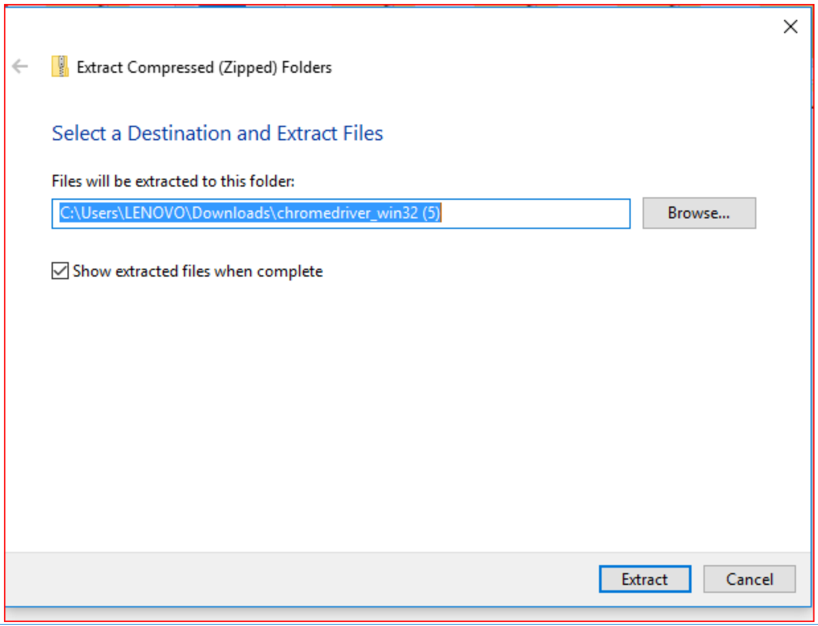
Current Releases

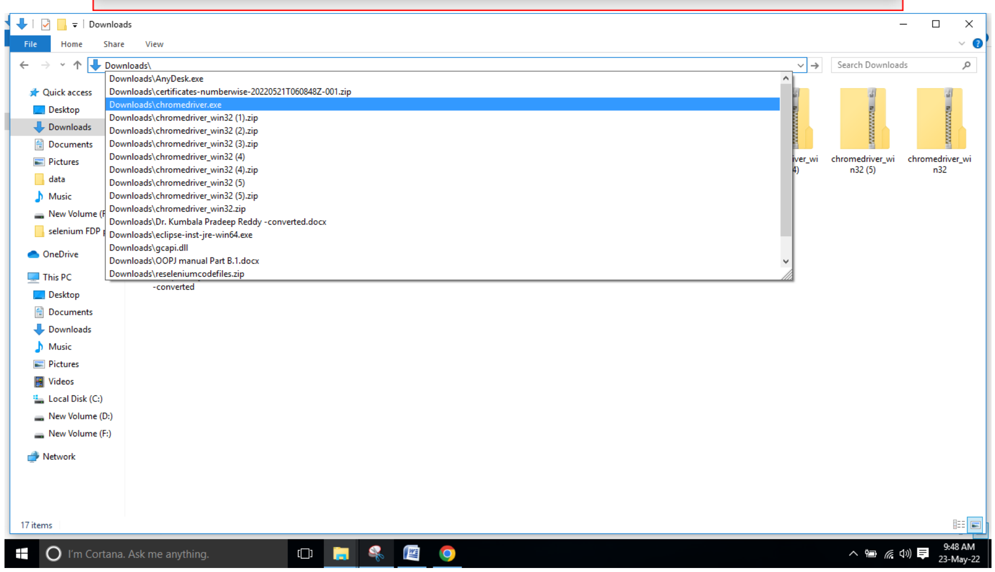
- If you are using Chrome version 102, please download ChromeDriver 102.0.5005.27
- If you are using Chrome version 101, please download ChromeDriver 101.0.4951.41
- If you are using Chrome version 100, please download ChromeDriver 100.0.4896.60


Index of /101.0.4951.41/
Name
Last modified
Size
ETag
Parent Directory
-
chromedriver_linux64.zip
2022-04-27 07:02:29
5.92MB
chromedriver_mac64.zip
2022-04-27 07:02:31
7.88MB
chromedriver_mac64_m1.zip
2022-04-27 07:02:34
7.19MB
chromedriver_win32.zip
2022-04-27 07:02:37
6.05MB
notes.txt
2022-04-27 07:02:42
0.00MB


chromedriver_win3....zip
3.7/6.1 MB, 6 secs left


chromedriver_wi
n32 (5)


Extract Compressed (Zipped) Folders
Select a Destination and Extract Files
Files will be extracted to this folder:
C:\Users\LENOVO\Downloads\chromedriver_win32 (5)
Browse...
Show extracted files when complete
Extract
Cancel


Downloads
File
Home
Share
View
Downloads\
Search Downloads
Downloads\AnyDesk.exe
Downloads\certificates-numberwise-20220521T060848Z-001.zip
Downloads\chromedriver.exe
Downloads\chromedriver_win32 (1).zip
Downloads\chromedriver_win32 (2).zip
Downloads\chromedriver_win32 (3).zip
Downloads\chromedriver_win32 (4)
Downloads\chromedriver_win32 (4).zip
Downloads\chromedriver_win32 (5)
Downloads\chromedriver_win32 (5).zip
Downloads\chromedriver_win32.zip
Downloads\Dr. Kumbala Pradeep Reddy -converted.docx
Downloads\eclipse-inst-jre-win64.exe
Downloads\gcapi.dll
Downloads\OOPJ manual Part B.1.docx
Downloads\reseleniumcodefiles.zip
chromedriver_win32 (5)
chromedriver_win32
Quick access
Desktop
Downloads
Documents
Pictures
data
Music
selenium FDP
OneDrive
This PC
Videos
Local Disk (C:)
New Volume (D:)
New Volume (F:)
Network
17 items
I'm Cortana. Ask me anything.
9:48 AM
23-May-22

WEEK-6

AIM:
Write a test case to perform automation on Ajio shopping website.

ALGORITHM:

1. Accessing shopping website
2. create an account for that Ajio shopping website.
3. Enter the fields through testcases information we have provided.
4. close the acivity after successfully created account.

SOURCE CODE:

```
package CMRIT.CMRIT_Assignment;

import java.util.Iterator;
import java.util.Set;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;

public class Ajio {

	public static void main(String[] args) throws Exception {

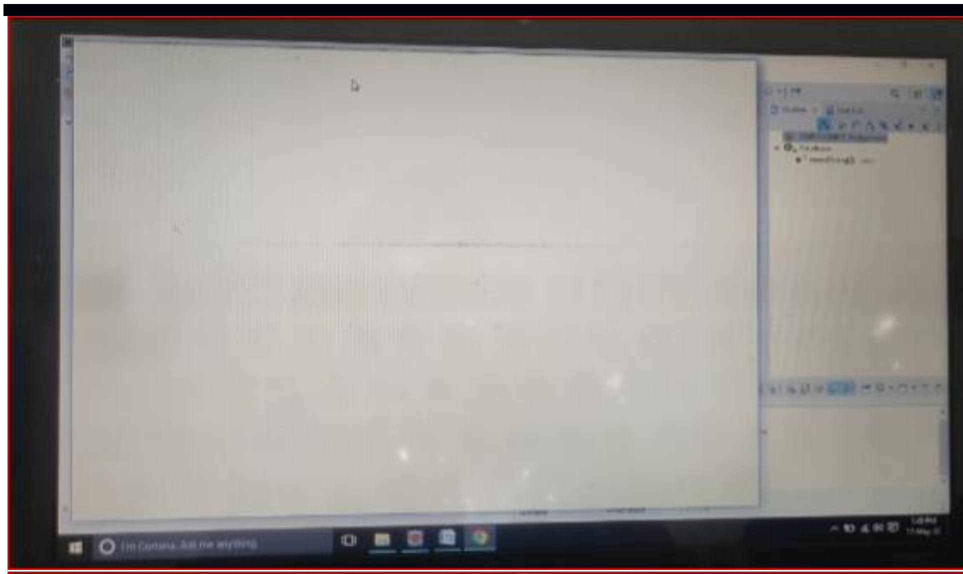
	System.setProperty("webdriver.chrome.driver","C:\\Users\\LENOVO\\Downloads\\chro
medriver.exe");
		WebDriver driver = new ChromeDriver();
		driver.get("https://www.ajio.com/");
		driver.manage().window().maximize();
		Thread.sleep(2000);
		WebElement ajioLink = driver.findElement(By.xpath("//span[normalize-
space()='Sign In / Join AJIO']"));
		ajioLink.click();
		Thread.sleep(2000);
		WebElement facebookBtn = driver.findElement(By.xpath("//span[normalize-
space()='Facebook']"));
		facebookBtn.click();
		Thread.sleep(2000);
		Set<String> parentWindow = driver.getWindowHandles();
		Iterator<String> iterator = parentWindow.iterator();
		while(iterator.hasNext())
			{
				String childWindow = iterator.next();
```

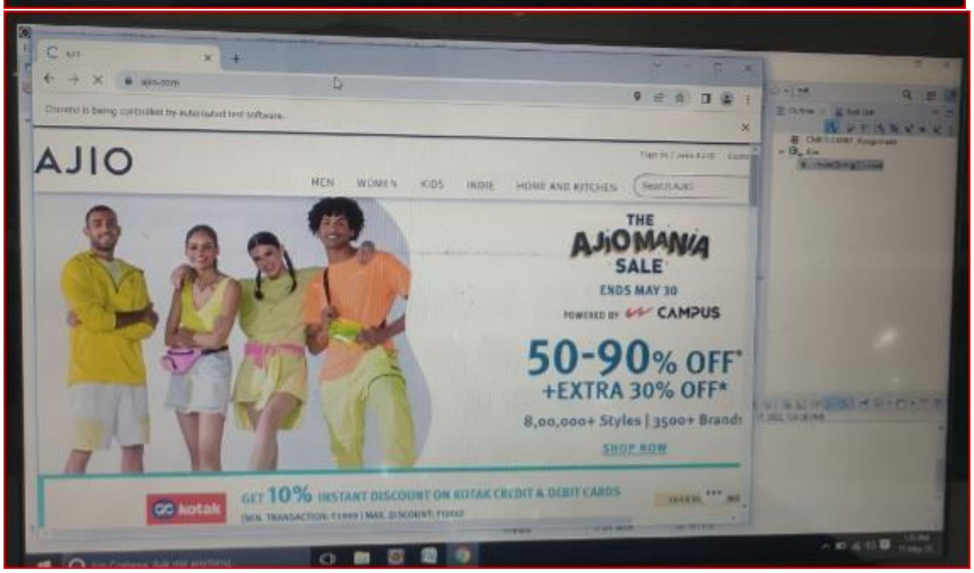
```
                    if(!parentWindow.equals(childWindow))
                    {
                        driver.switchTo().window(childWindow);
                    }
                }
            WebElement emailOrMobileNo =
driver.findElement(By.xpath("//input[@id='email']"));
            emailOrMobileNo.sendKeys("martin@gmail.com");
            WebElement pwd = driver.findElement(By.xpath("//input[@id='pass']"));
            pwd.sendKeys("Martin@123");
            WebElement loginBtn = driver.findElement(By.cssSelector("body > div:nth-
child(2) > div:nth-child(2) > div:nth-child(2) > form:nth-child(1) > div:nth-child(4) > div:nth-
child(16) > label:nth-child(2) > input:nth-child(1)"));
            loginBtn.click();

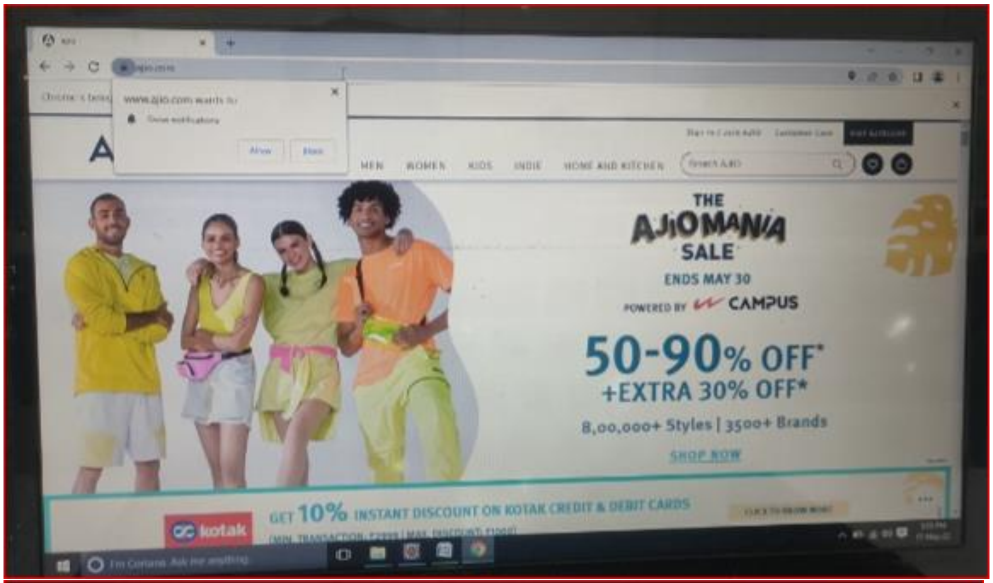
        }

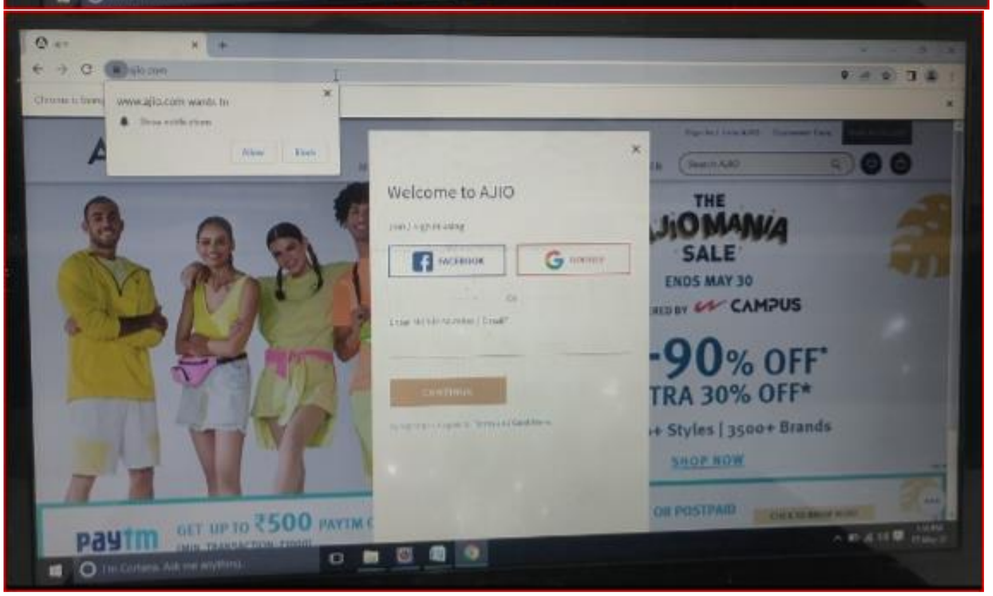
}
```

OUTPUT :




Chrome is being controlled by automated test software.
AJIO
MEN
WOMEN
KIDS
INDIE
HOME AND KITCHEN
THE
AJIOMANIA
SALE
ENDS MAY 30
POWERED BY CAMPUS
50-90% OFF*
+EXTRA 30% OFF*
SHOP NOW
kotak
GET 10% INSTANT DISCOUNT ON KOTAK CREDIT & DEBIT CARDS


www.ajio.com wants to
MEN WOMEN KIDS INDIE HOME AND KITCHEN
THE
AJIOMANIA
SALE
ENDS MAY 30
POWERED BY CAMPUS
50-90% OFF*
+EXTRA 30% OFF*
8,00,000+ Styles | 3500+ Brands
SHOP NOW
kotak
GET 10% INSTANT DISCOUNT ON KOTAK CREDIT & DEBIT CARDS


www.ajio.com wants to
Welcome to AJIO
FACEBOOK
THE
SALE
ENDS MAY 30
CAMPUS
-90% OFF*
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SHOP NOW
Paytm
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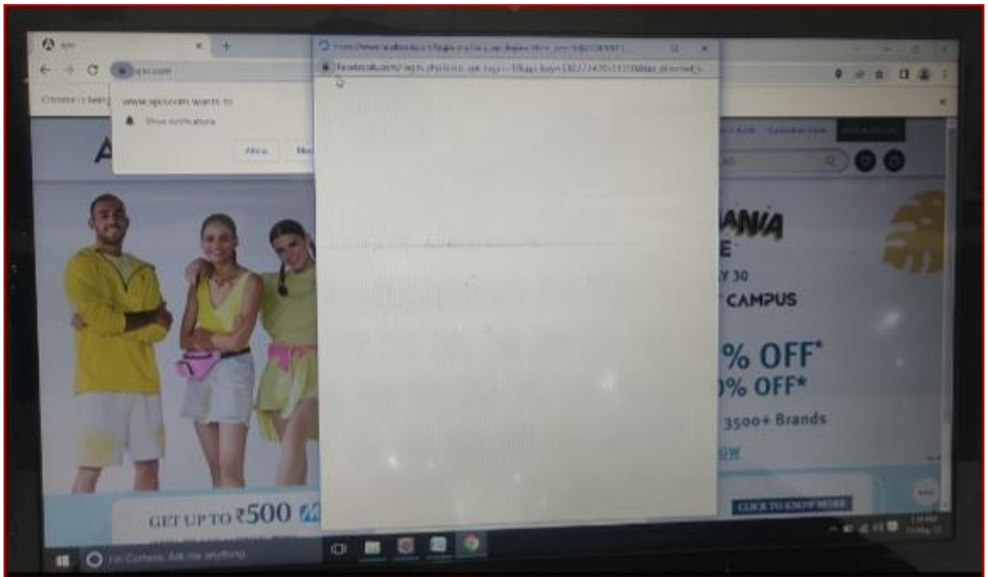
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Show notifications
CAMPUS
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3500+ Brands
CLICK TO KNOW MORE
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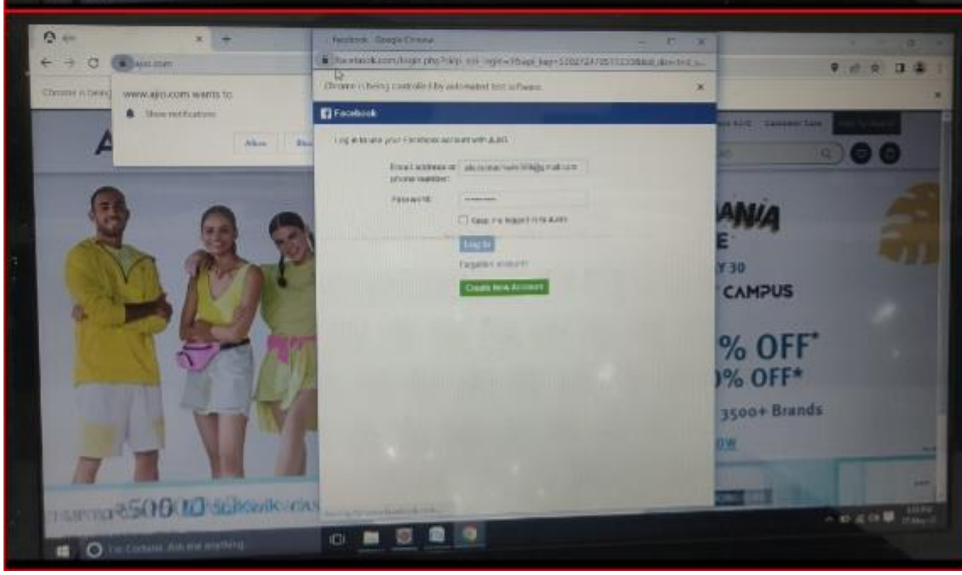

www.ajio.com wants to
Show notifications
Facebook
Log In
Create New Account
CAMPUS
% OFF*
% OFF*
3500+ Brands

Viva Questions:

1) How you can use "submit" a form using Selenium ?

You can use "submit" method on element to submit form-

element.submit () ;

Alternatively you can use click method on the element which does form submission

2) What are the features of TestNG and list some of the functionality in TestNG which makes it more effective?

TestNG is a testing framework based on JUnit and NUnit to simplify a broad range of testing needs, from Unit Testing to Integration Testing. And the functionality which makes it efficient testing framework are

- Support for annotations
- Support for data-driven testing
- Flexible test configuration
- Ability to re-execute failed test cases

3) Explain what is the difference between find elements () and find element () ?

find element ():

It finds the first element within the current page using the given "locating mechanism". It returns a single WebElement

findElements () : Using the given "locating mechanism" find all the elements within the current page. It returns a list of web elements.

4) Which attribute you should consider throughout the script in frame for "if no frame Id as well as no frame name"?

You can use…..driver.findElements(By.xpath("//iframe"))….

This will return list of frames.

You will need to switch to each and every frame and search for locator which we want.

Then break the loop

5) Explain what are the JUnits annotation linked with Selenium?

The JUnits annotation linked with Selenium are

- @Before public void method() – It will perform the method () before each test, this method can prepare the test

- @Test public void method() – Annotations @Test identifies that this method is a test method environment
- @After public void method()- To execute a method before this annotation is used, test method must start with test@Before

WEEK-6

AIM:
Write a test case to perform automation on Ajio shopping website.

ALGORITHM:

5. Accessing shopping website
6. create an account for that Ajio shopping website.
7. Enter the fields through testcases information we have provided.
8. close the acivity after successfully created account.

SOURCE CODE:

```
package CMRIT.CMRIT_Assignment;

import java.util.Iterator;
import java.util.Set;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;

public class Ajio {

	public static void main(String[] args) throws Exception {

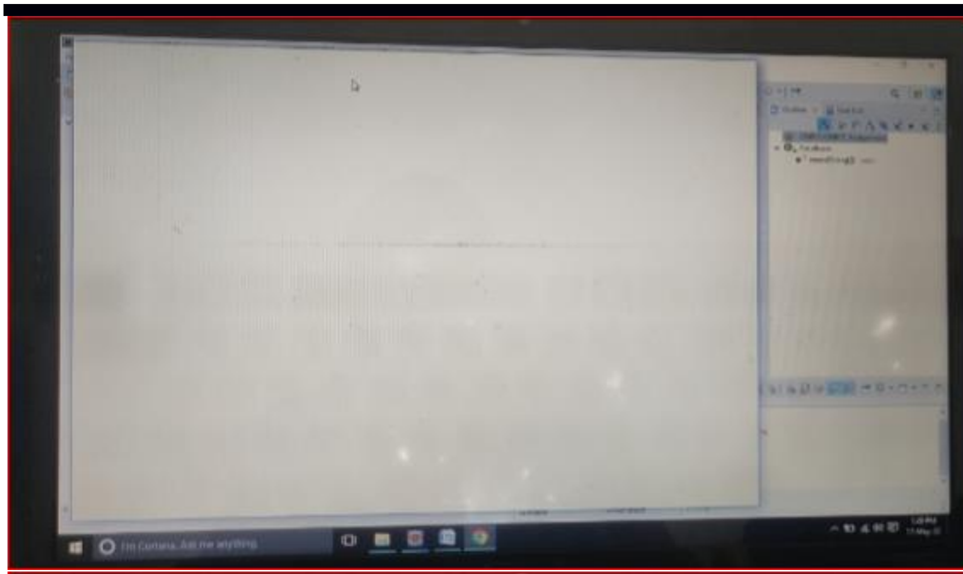
	System.setProperty("webdriver.chrome.driver","C:\\Users\\LENOVO\\Downloads\\chro
medriver.exe");
		WebDriver driver = new ChromeDriver();
		driver.get("https://www.ajio.com/");
		driver.manage().window().maximize();
		Thread.sleep(2000);
		WebElement ajioLink = driver.findElement(By.xpath("//span[normalize-
space()='Sign In / Join AJIO']"));
		ajioLink.click();
		Thread.sleep(2000);
		WebElement facebookBtn = driver.findElement(By.xpath("//span[normalize-
space()='Facebook']"));
		facebookBtn.click();
		Thread.sleep(2000);
		Set<String> parentWindow = driver.getWindowHandles();
		Iterator<String> iterator = parentWindow.iterator();
		while(iterator.hasNext())
			{
				String childWindow = iterator.next();
```

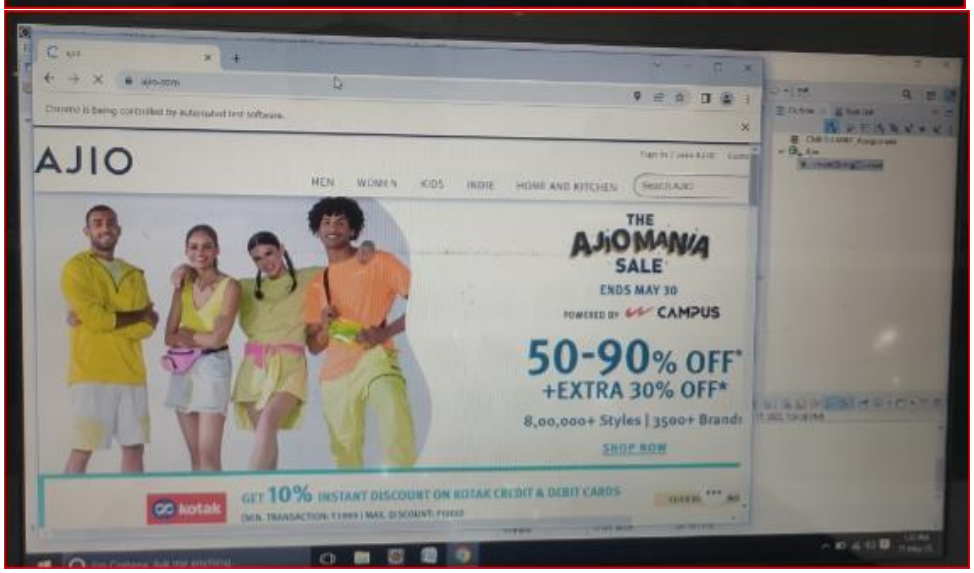
```
                        if(!parentWindow.equals(childWindow))
                        {
                            driver.switchTo().window(childWindow);
                        }
                    }
                WebElement emailOrMobileNo =
driver.findElement(By.xpath("//input[@id='email']"));
                emailOrMobileNo.sendKeys("martin@gmail.com");
                WebElement pwd = driver.findElement(By.xpath("//input[@id='pass']"));
                pwd.sendKeys("Martin@123");
                WebElement loginBtn = driver.findElement(By.cssSelector("body > div:nth-
child(2) > div:nth-child(2) > div:nth-child(2) > form:nth-child(1) > div:nth-child(4) > div:nth-
child(16) > label:nth-child(2) > input:nth-child(1)"));
                loginBtn.click();

        }

}
```

OUTPUT :




Chrome is being controlled by automated test software.
AJIO
MEN
WOMEN
KIDS
INDIE
HOME AND KITCHEN
THE
AJIOMANIA
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+EXTRA 30% OFF*
8,00,000+ Styles | 3500+ Brand
SHOP NOW
kotak
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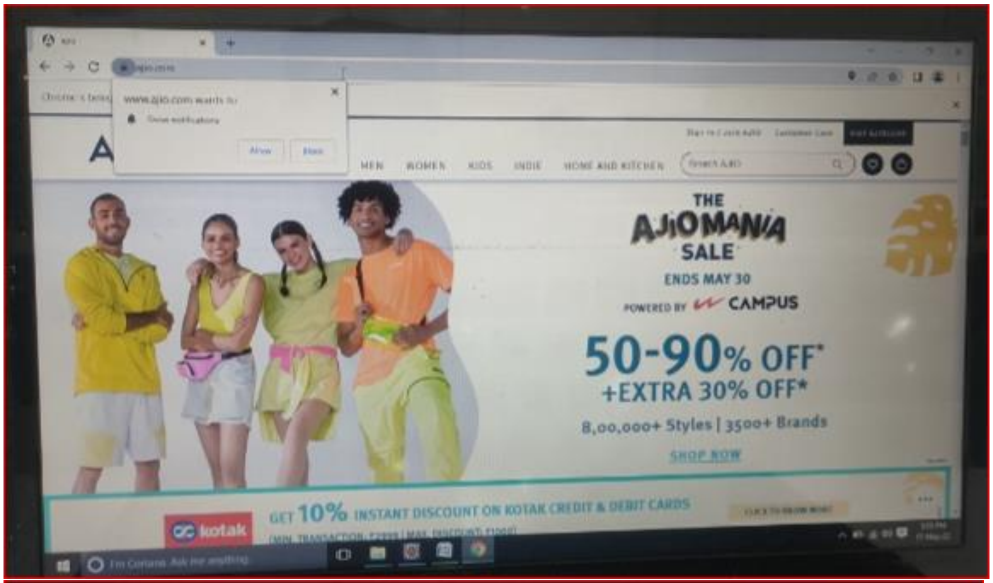
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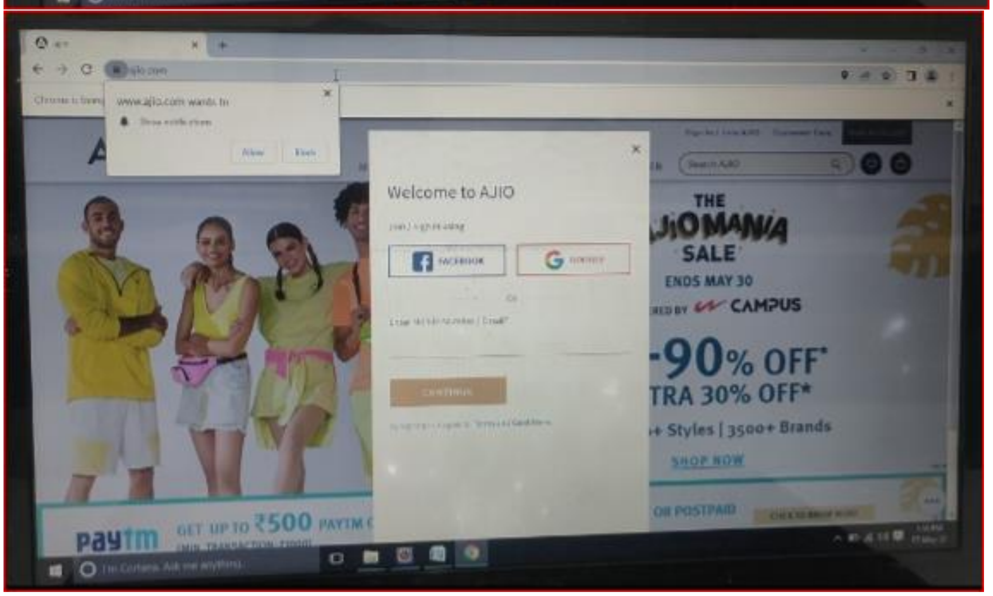
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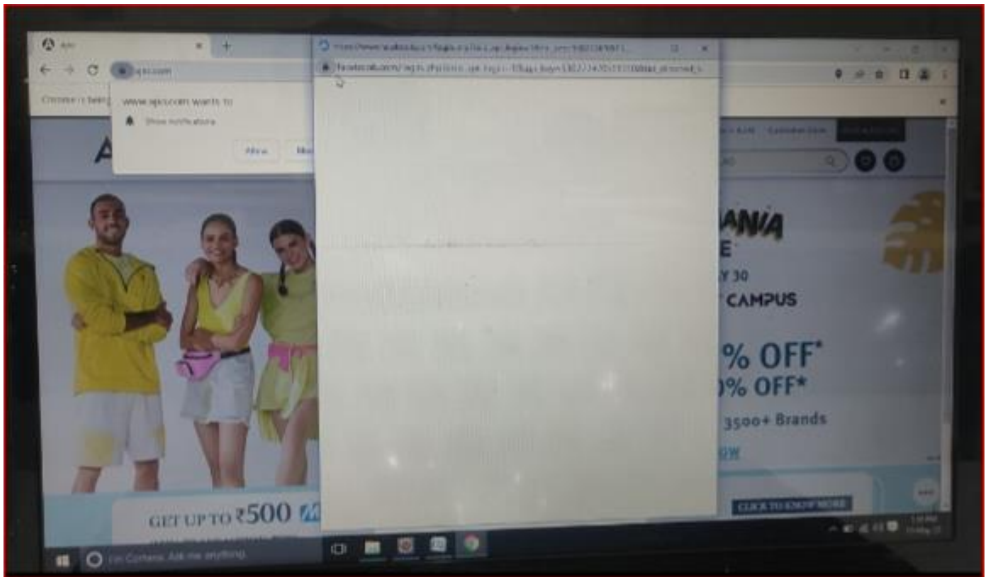
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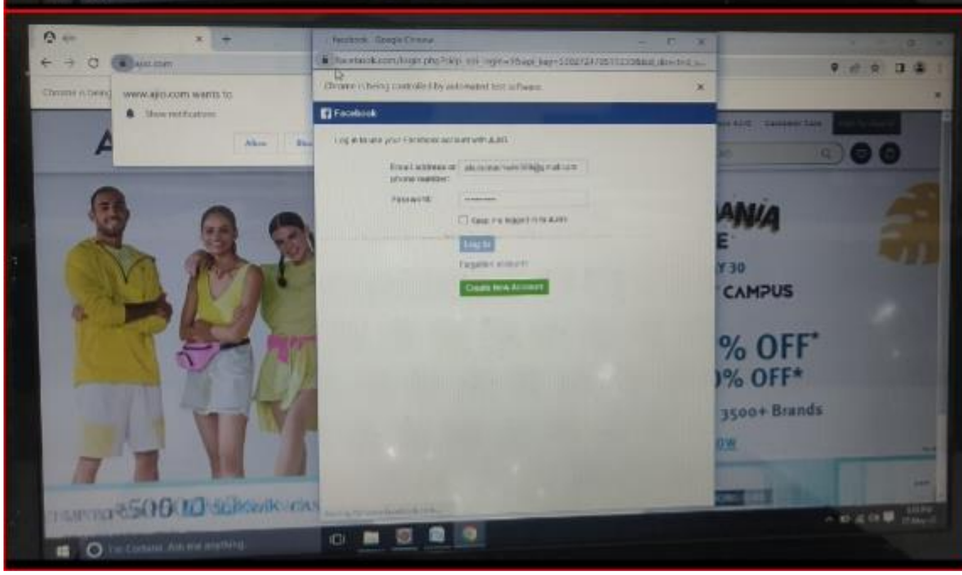
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WEEK-1

AIM:
Write a program in web driver to open Google and search CMRIT.

ALGORITHM:

1. Search google website
2. Fetching the google search bar
3. Search CMRIT
4. Exit from page.

SOURCE CODE:

```
package CMRIT.CMRIT_Assignment;

import org.openqa.selenium.By;
import org.openqa.selenium.Keys;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;

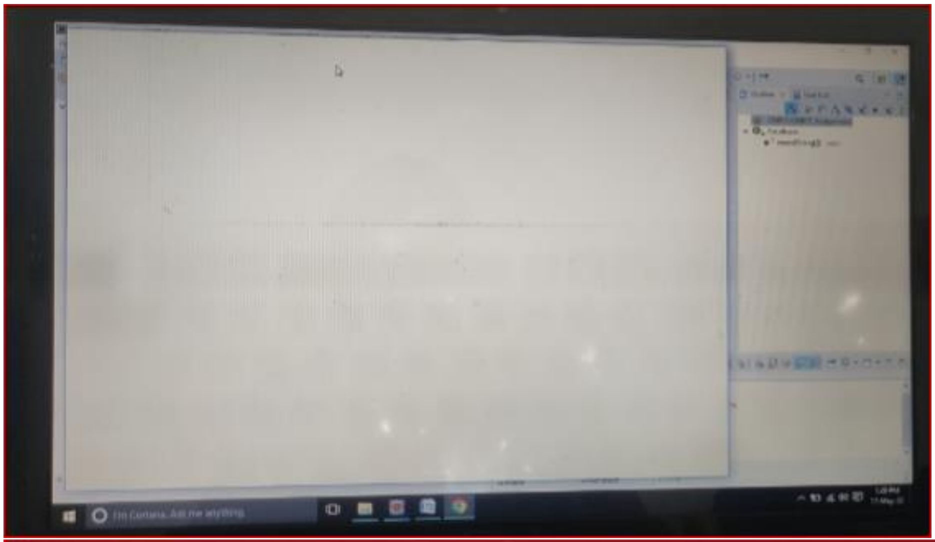
public class GoogleSearch_CMRIT {

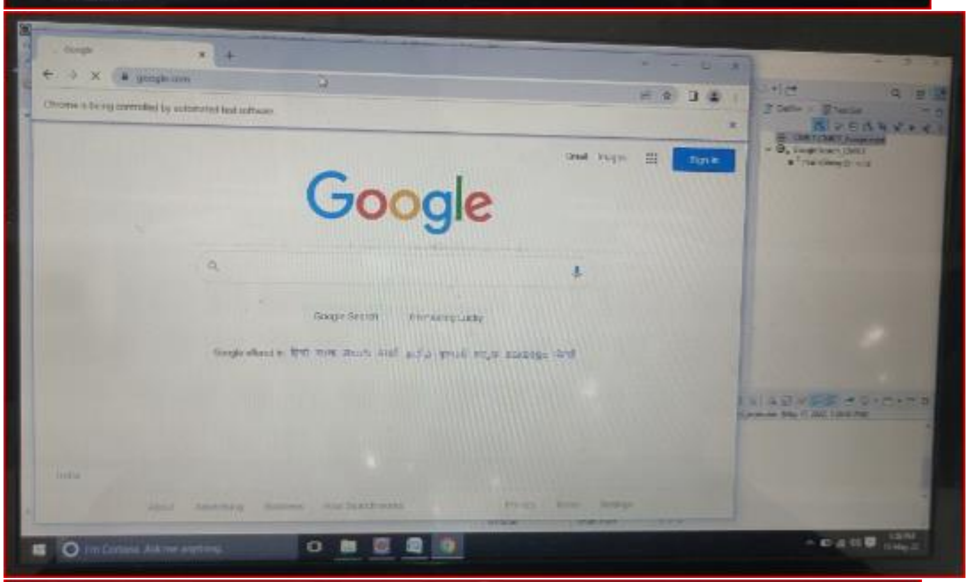
        public static void main(String[] args) throws InterruptedException
        {
System.setProperty("webdriver.chrome.driver",(" C:\Users\hp\Downloads\chromedriver-
win64");
driver = new ChromeDriver();
                driver.get("https://www.google.com/");
                driver.manage().window().maximize();
                Thread.sleep(2000);
                WebElement searchBar = driver.findElement(By.name("q"));
                searchBar.sendKeys("CMRIT");
                searchBar.sendKeys(Keys.ENTER);
                Thread.sleep(15000);
                driver.quit();

        }

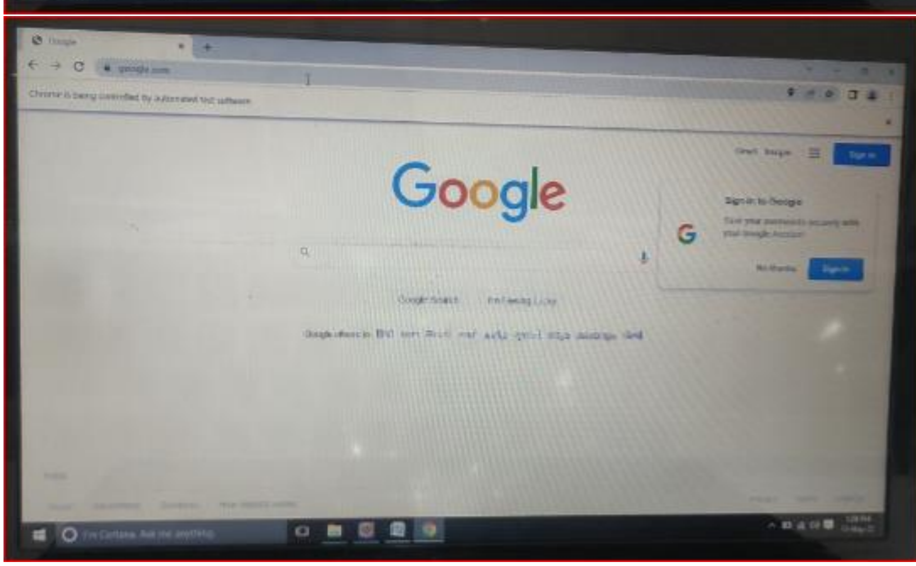
}
```

OUTPUT:

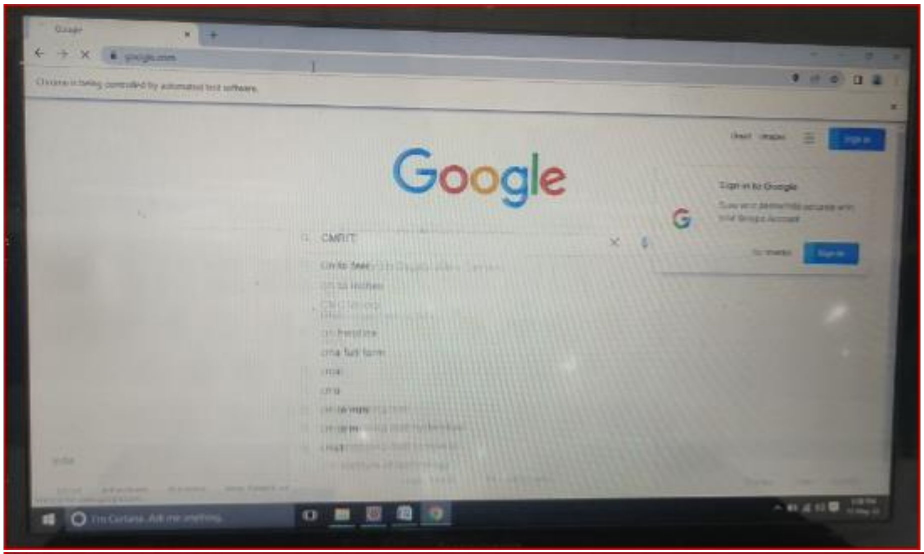


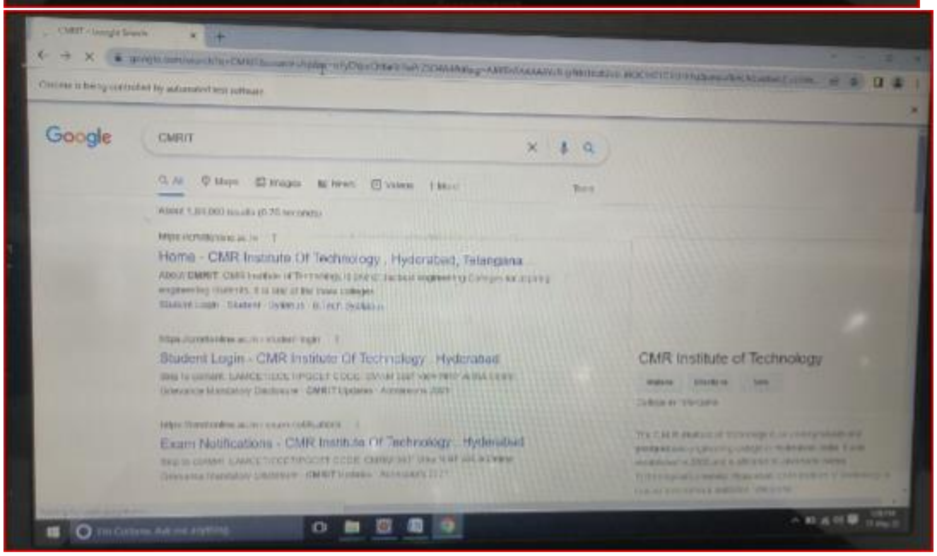


Chrome is being controlled by automated test software
Google



Chrome is being controlled by automated test software
Google


Google
CMRIT


Google
CMRIT
Home - CMR Institute Of Technology , Hyderabad, Telangana
Student Login - CMR Institute Of Technology , Hyderabad
Exam Notifications - CMR Institute Of Technology , Hyderabad
CMR Institute of Technology

WEEK-2

AIM:
Write a test case to search result section on CMRIT Website.

ALGORITHM:

1. Open google website
2. Search for CMRIT site.
3. Take one official ID of any student in cmrit college.
4. By using that Id we are accessing CMRIT website and giving the info of that student
5. It will redirected to opening the results of that student.

SOURCE CODE:

```
package CMRIT.CMRIT_Assignment;

import org.openqa.selenium.By;
import org.openqa.selenium.Keys;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.interactions.Actions;

public class CMRIT_StudentResults {

        public static void main(String[] args) throws InterruptedException
        {
System.setProperty("webdriver.chrome.driver",(" C:\Users\hp\Downloads\chromedriver-win64")
        ;       WebDriver driver = new ChromeDriver();
                driver.manage().window().maximize();
                driver.get("https://www.google.com/");

                Thread.sleep(2000);
                WebElement searchBar = driver.findElement(By.name("q"));
                searchBar.sendKeys("CMRIT");
                searchBar.sendKeys(Keys.ENTER);

                WebElement cmritLink =
driver.findElement(By.xpath("(//h3[contains(text(),'Home - CMR Institute Of Technology ,
Hyderabad, Te')])[1]"));
                cmritLink.click();

                Thread.sleep(3000);
```

```
		Actions builder = new Actions(driver);

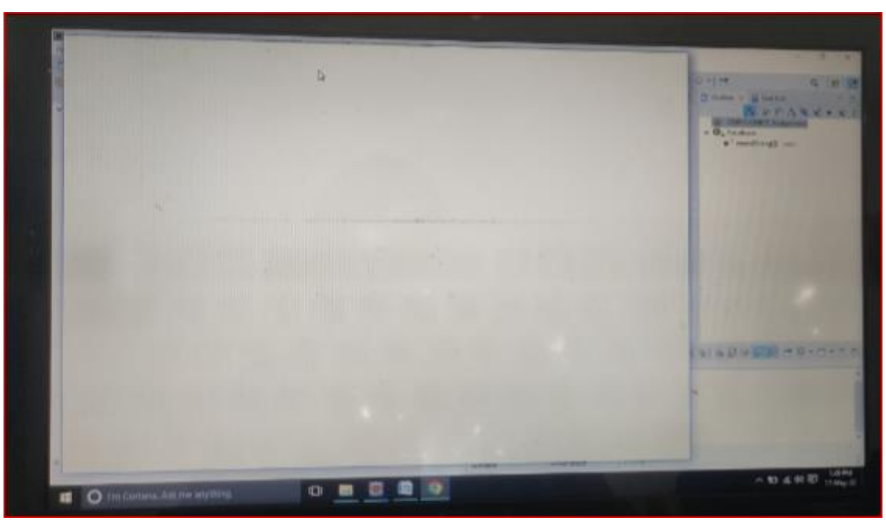
		WebElement student =
driver.findElement(By.xpath("/html[1]/body[1]/div[1]/div[1]/section[3]/div[1]/div[1]/div[1]/div[1]/div[1]/nav[1]/ul[1]/li[9]/a[1]"));
		builder.moveToElement(student).perform();
		student.click();
		Thread.sleep(2000);
		WebElement studentLogin =
driver.findElement(By.xpath("/html[1]/body[1]/div[1]/div[1]/section[3]/div[1]/div[1]/div[1]/div[1]/div[1]/nav[1]/ul[1]/li[9]/ul[1]/li[1]/a[1]"));
		studentLogin.click();
		WebElement userName =
driver.findElement(By.xpath("(//input[@id='txtUserName'])[1]"));
		Thread.sleep(2000);
		userName.sendKeys("20R01A05K6p");

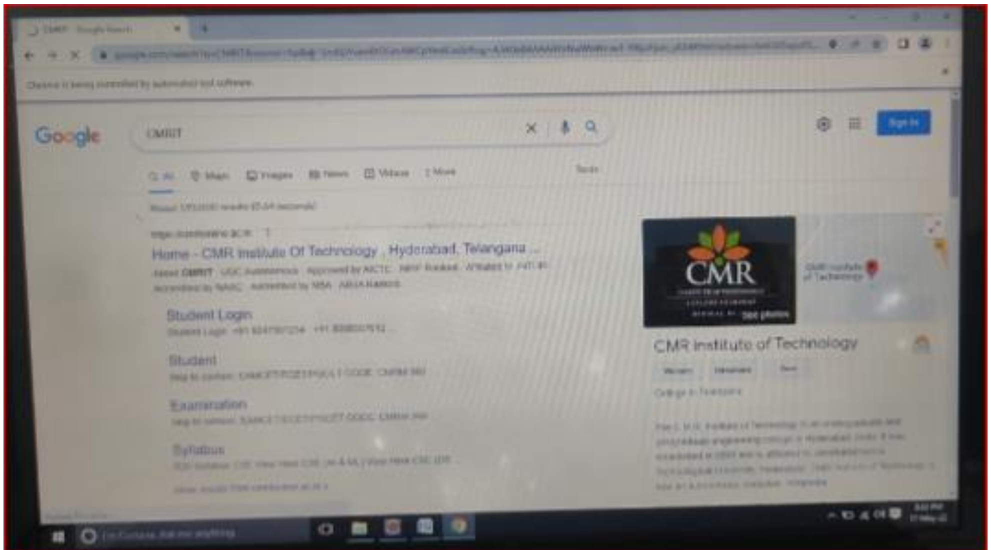
		WebElement nxtBtn = driver.findElement(By.name("btnNext"));
		nxtBtn.click();

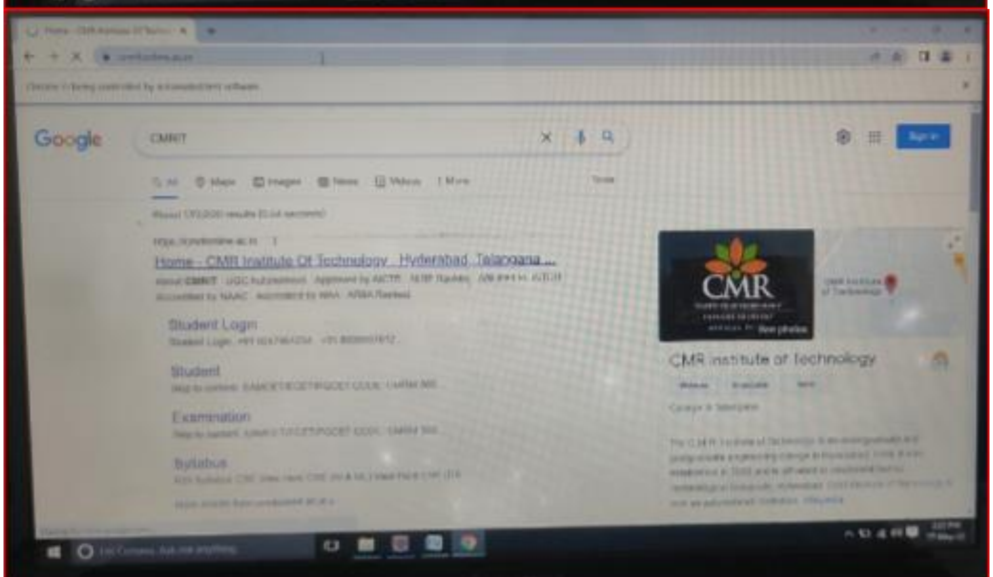
		WebElement password =
driver.findElement(By.xpath("(//input[@placeholder='Enter Password'])[1]"));
		password.sendKeys("20R01A05K6P");

		WebElement submit =
driver.findElement(By.xpath("(//input[@name='btnSubmit'])[1]"));
		submit.click();
		}
}
```

OUTPUT:




Google
Home - CMR Institute Of Technology , Hyderabad, Telangana ...
Student Login
Student
Examination
Syllabus
CMR
CMR Institute of Technology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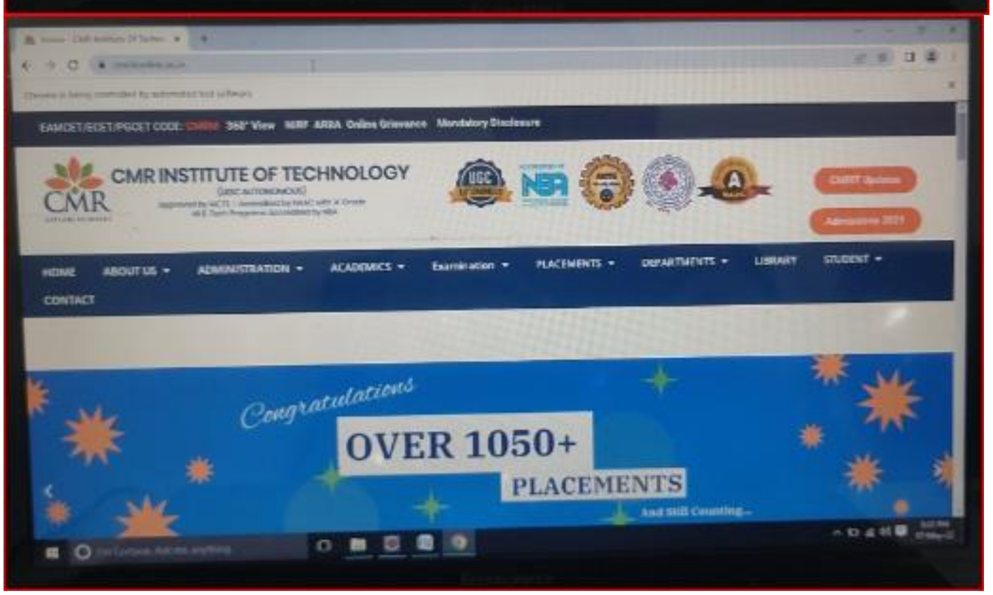
EAMCET/ECET/PGECET CODE:
360° View
NIRF
ARIIA
Online Grievance
Mandatory Disclosure
CMR INSTITUTE OF TECHNOLOGY
HOME
ABOUT US
ADMINISTRATION
ACADEMICS
Examination
PLACEMENTS
DEPARTMENTS
LIBRARY
STUDENT
CONTACT
Congratulations
OVER 1050+
PLACEMENTS
And Still Counting...


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ABOUT US
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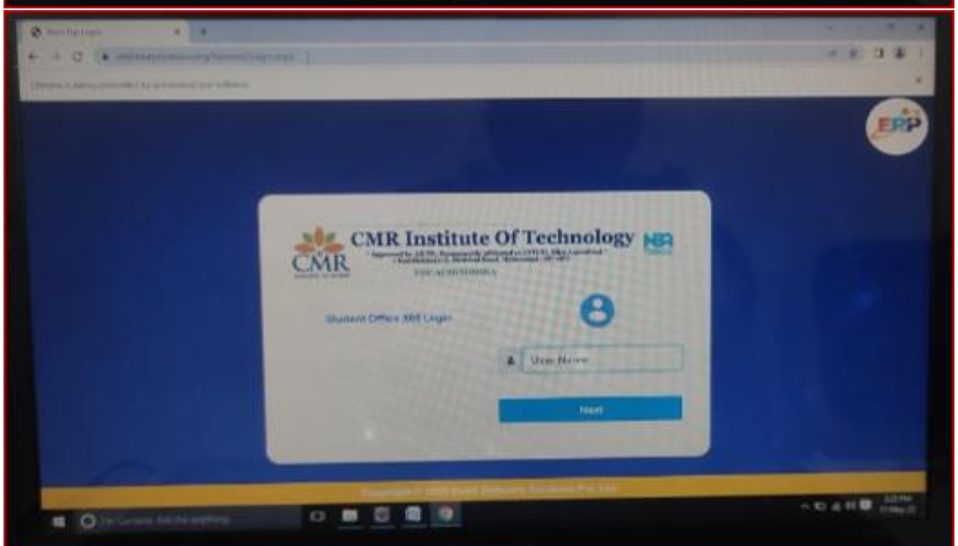
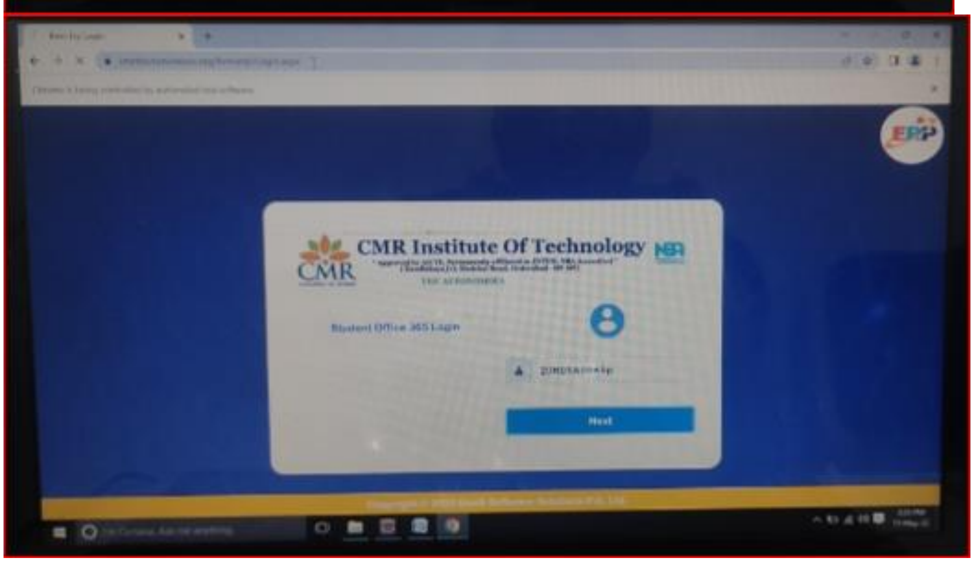
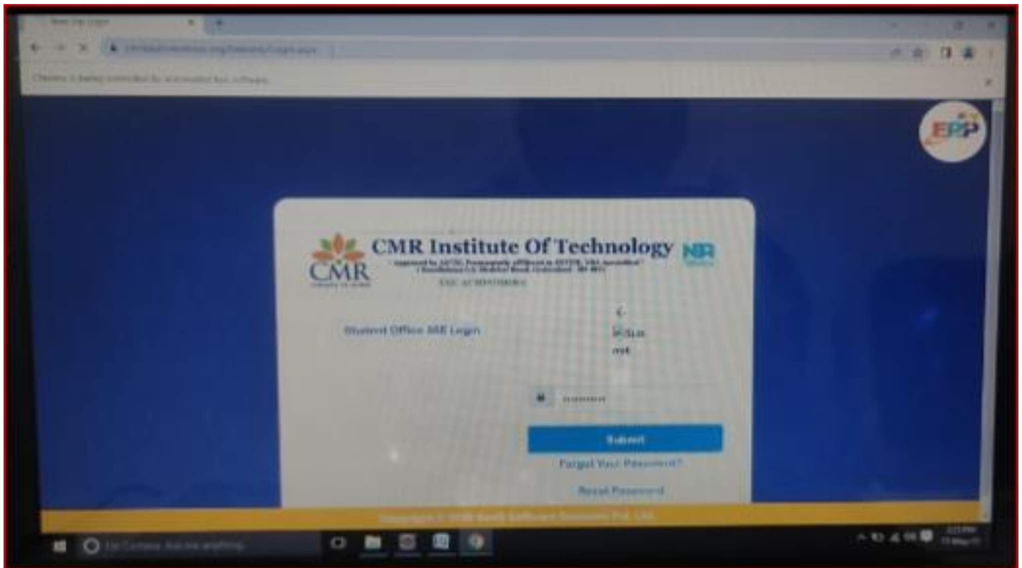
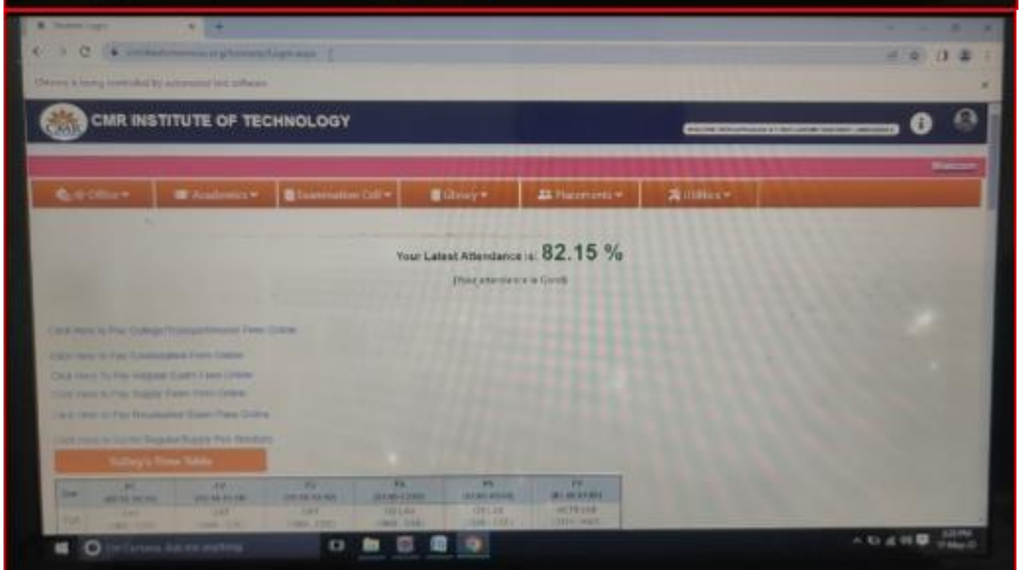
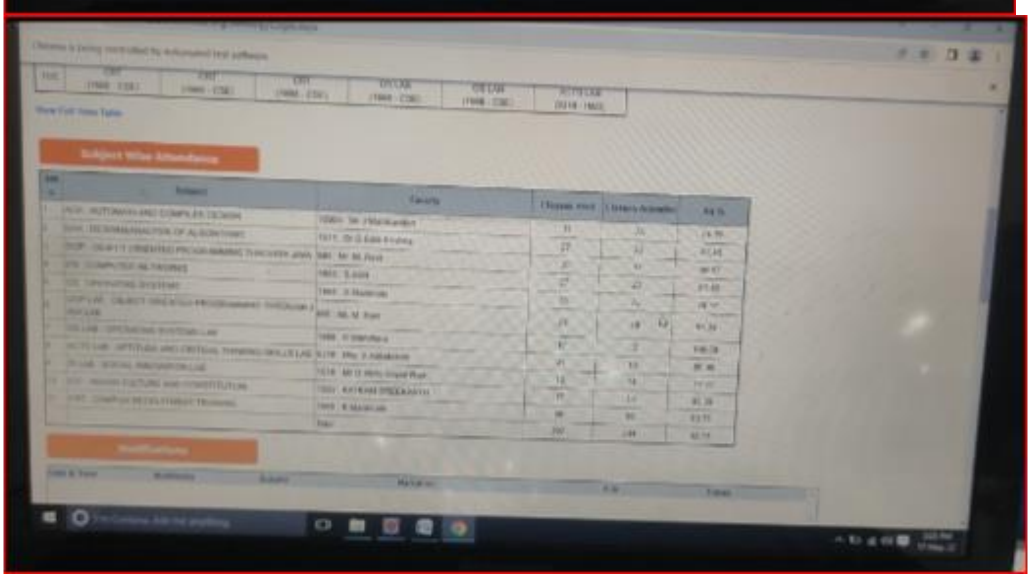

CMR Institute Of Technology
Student Office 365 Login
User Name
Next


CMR Institute Of Technology
Student Office 365 Login
Next


CMR Institute Of Technology
CMR
Student Office MIS Login
Submit
Forgot Your Password?
Reset Password


CMR INSTITUTE OF TECHNOLOGY
Your Latest Attendance is: 82.15 %
Today's Time Table


Subject Wise Attendance
Notifications

WEEK- 3

AIM:
Write test case to open google and download a image from google images of cmrit website.

ALGORITHM:

1. search google website
2. Fetching the google searchBar
3. Sending CMRIT in searchbar
4. Opening the images page
5. Select one image and download that image into our system downloads.

SOURCE CODE:

```
package CMRIT.CMRIT_Assignment;

import java.awt.Robot;
import java.awt.event.KeyEvent;

import org.openqa.selenium.By;
import org.openqa.selenium.Keys;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.interactions.Actions;

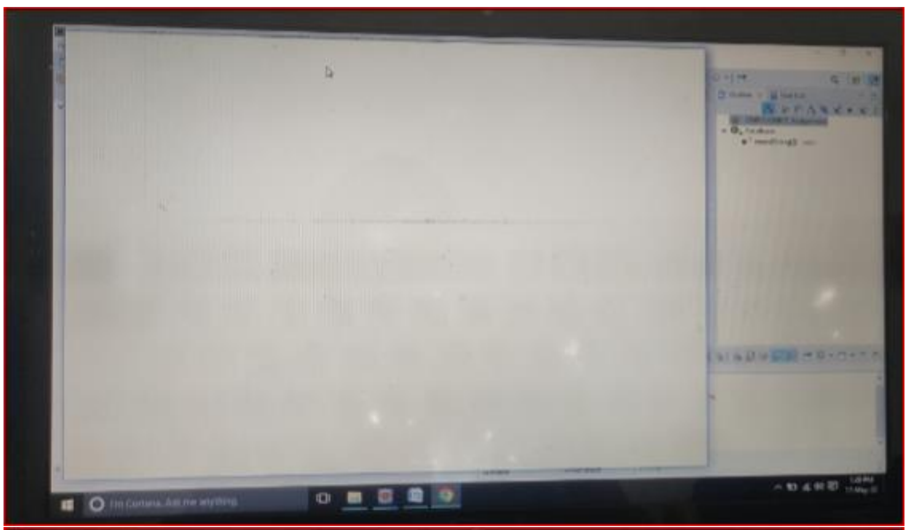
public class GoogleImageDownload {
	public static void main(String[] args) throws Exception {
		System.setProperty("webdriver.chrome.driver"," C:\Users\hp\Downloads\chromedriver-win64 ");
		WebDriver driver = new ChromeDriver();
		driver.manage().window().maximize();
		driver.get("https://www.google.com/");
		Thread.sleep(2000);
		WebElement searchBar = driver.findElement(By.name("q"));
		searchBar.sendKeys("CMRIT");
		searchBar.sendKeys(Keys.ENTER);
		Thread.sleep(2000);
		driver.findElement(By.xpath("//a[normalize-space()='Images']")).click();

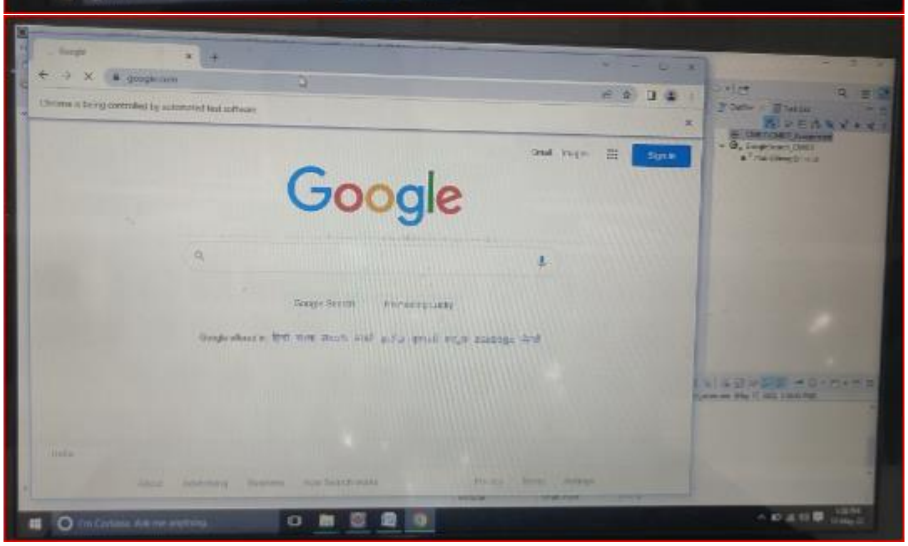
		WebElement Image = driver.findElement(By.xpath("//img[@alt='The CMRIT Campus - CMR
IT']"));
		Actions action = new Actions(driver);
		action.contextClick(Image).build().perform();
		Robot robot = new Robot();
		robot.keyPress(KeyEvent.VK_DOWN);
		Thread.sleep(500);
		robot.keyPress(KeyEvent.VK_DOWN);
		Thread.sleep(500);
		robot.keyPress(KeyEvent.VK_DOWN);
```

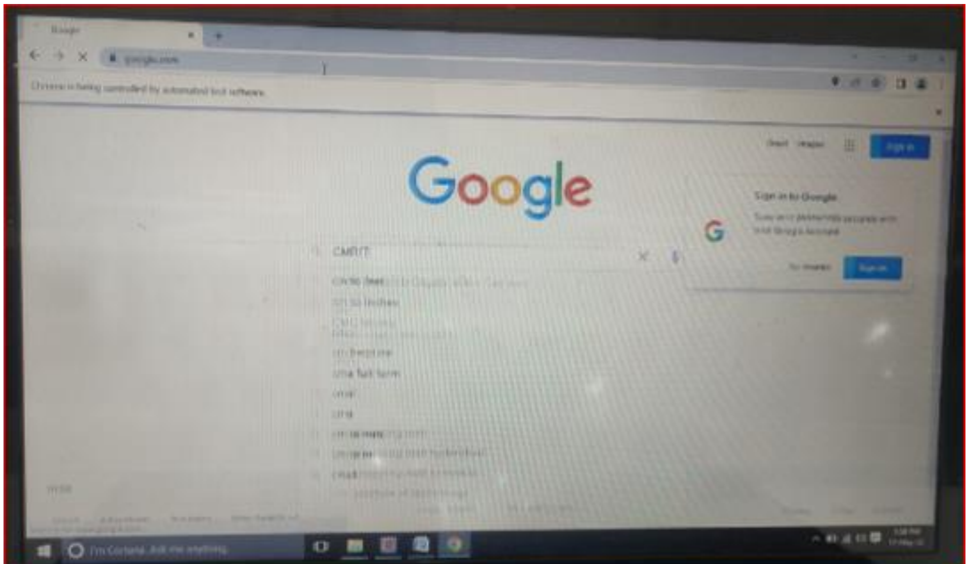
```
			Thread.sleep(500);
			robot.keyPress(KeyEvent.VK_DOWN);
			Thread.sleep(500);
			robot.keyPress(KeyEvent.VK_DOWN);
			Thread.sleep(500);
			robot.keyPress(KeyEvent.VK_DOWN);
			Thread.sleep(500);
			robot.keyPress(KeyEvent.VK_DOWN);
			Thread.sleep(500);
			robot.keyPress(KeyEvent.VK_DOWN);
			Thread.sleep(500);
			robot.keyPress(KeyEvent.VK_ENTER);
			Thread.sleep(3000);
			robot.keyPress(KeyEvent.VK_ENTER);
			System.out.println("downloaded");
			driver.close();

}
}
```

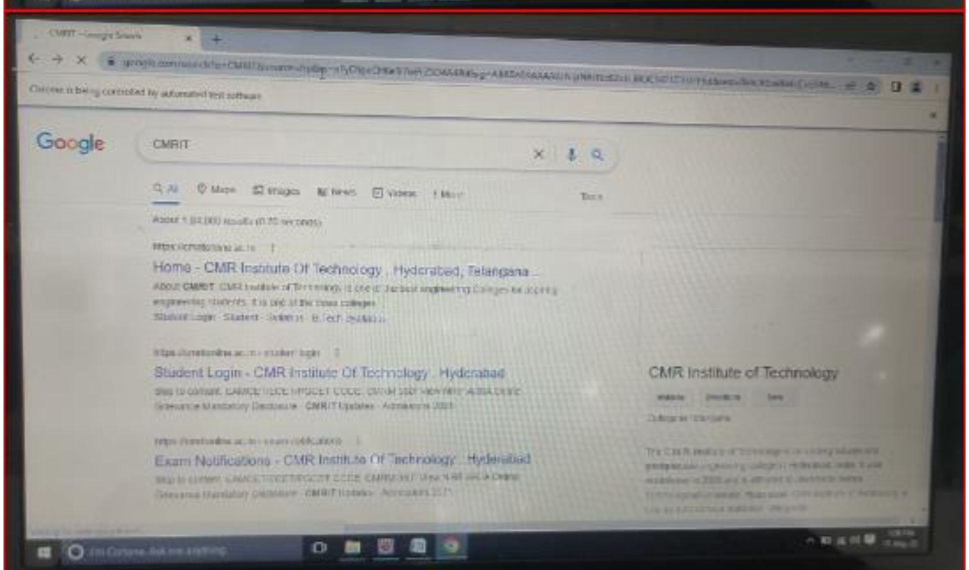
OUTPUT:







Google



Google
CMRIT
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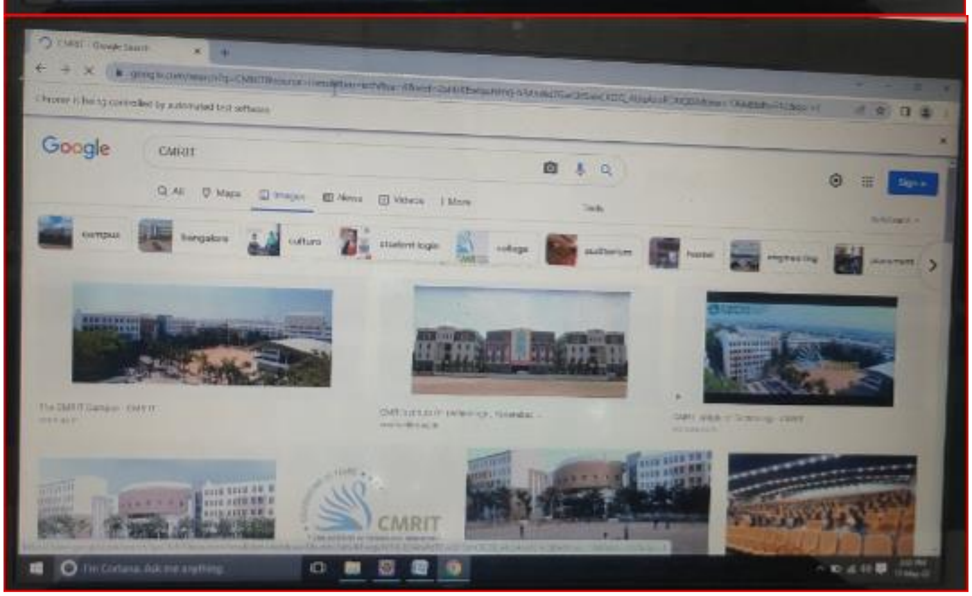
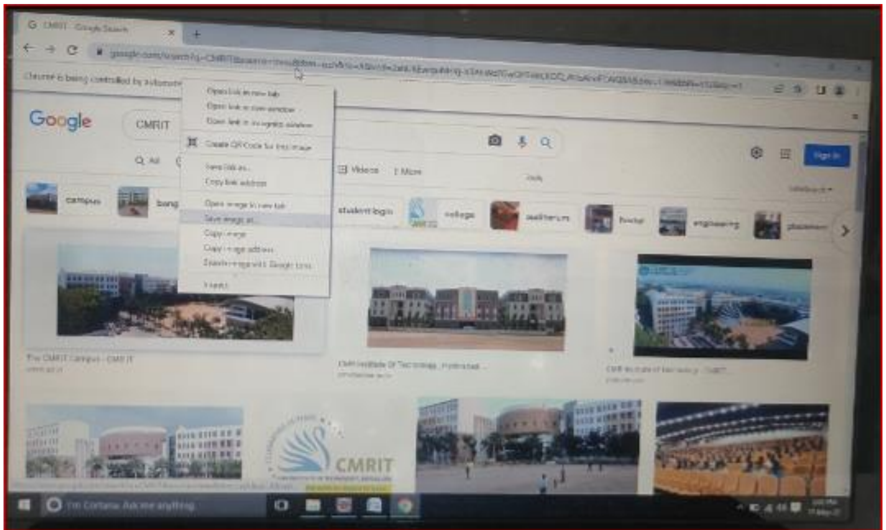
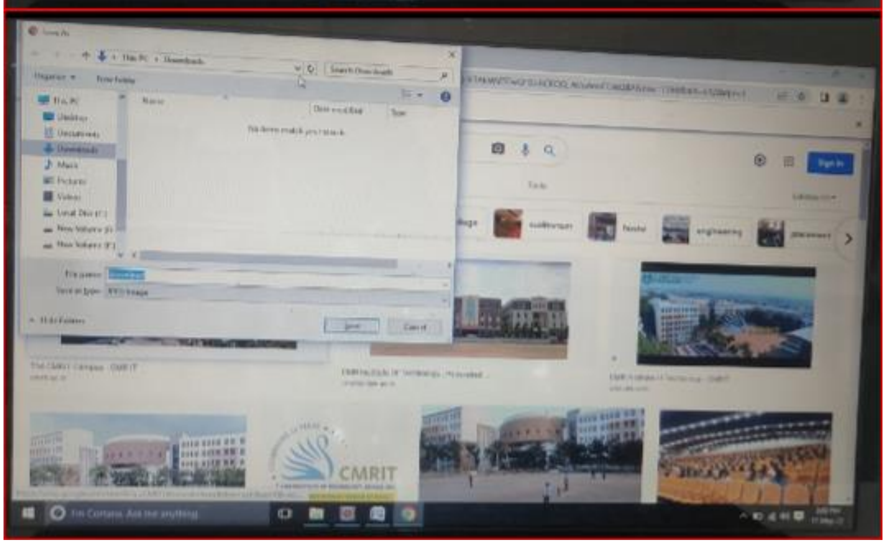
CMR Institute of Technology



Google
CMRIT


Google
CMRIT
Sign in
campus
student login
college
auditorium
hostel
engineering
CMRIT
I'm Cortana. Ask me anything

Sign in
auditorium
hostel
engineering
CMRIT
I'm Cortana. Ask me anything

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import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;

public class Ajio {

	public static void main(String[] args) throws Exception {

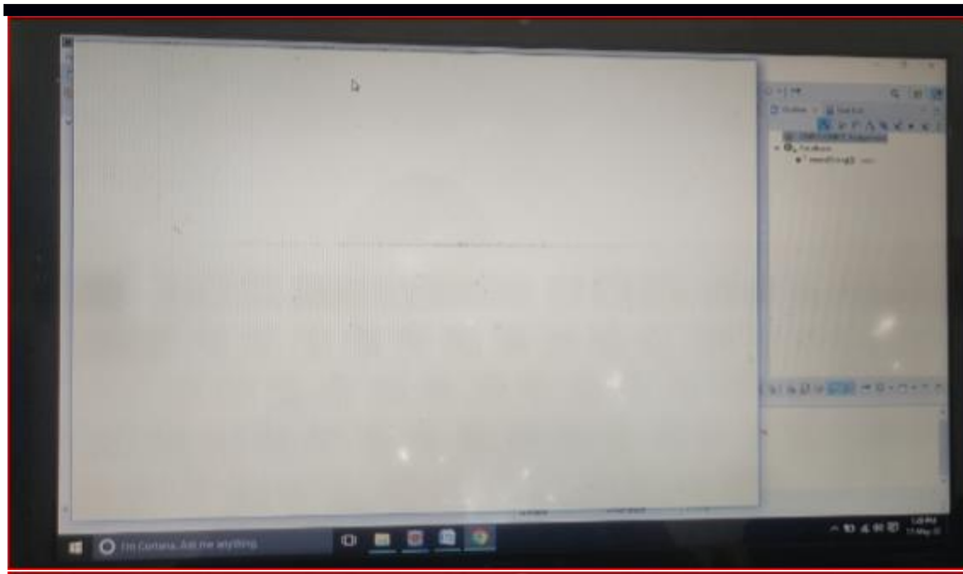
	System.setProperty("webdriver.chrome.driver","C:\\Users\\LENOVO\\Downloads\\chro
medriver.exe");
		WebDriver driver = new ChromeDriver();
		driver.get("https://www.ajio.com/");
		driver.manage().window().maximize();
		Thread.sleep(2000);
		WebElement ajioLink = driver.findElement(By.xpath("//span[normalize-
space()='Sign In / Join AJIO']"));
		ajioLink.click();
		Thread.sleep(2000);
		WebElement facebookBtn = driver.findElement(By.xpath("//span[normalize-
space()='Facebook']"));
		facebookBtn.click();
		Thread.sleep(2000);
		Set<String> parentWindow = driver.getWindowHandles();
		Iterator<String> iterator = parentWindow.iterator();
		while(iterator.hasNext())
			{
				String childWindow = iterator.next();
```

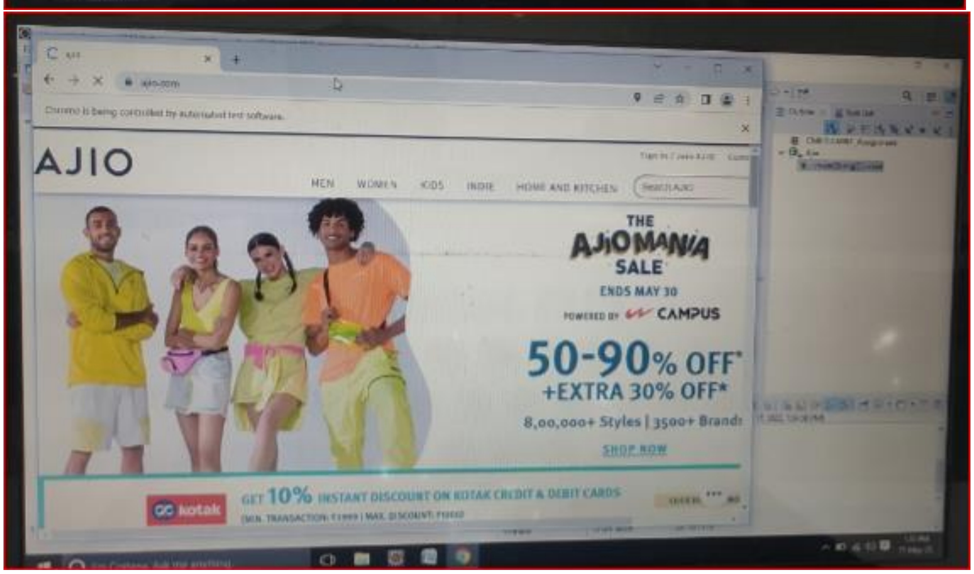
```
            if(!parentWindow.equals(childWindow))
            {
                driver.switchTo().window(childWindow);
            }
        }
    WebElement emailOrMobileNo =
driver.findElement(By.xpath("//input[@id='email']"));
    emailOrMobileNo.sendKeys("martin@gmail.com");
    WebElement pwd = driver.findElement(By.xpath("//input[@id='pass']"));
    pwd.sendKeys("Martin@123");
    WebElement loginBtn = driver.findElement(By.cssSelector("body > div:nth-
child(2) > div:nth-child(2) > div:nth-child(2) > form:nth-child(1) > div:nth-child(4) > div:nth-
child(16) > label:nth-child(2) > input:nth-child(1)"));
    loginBtn.click();

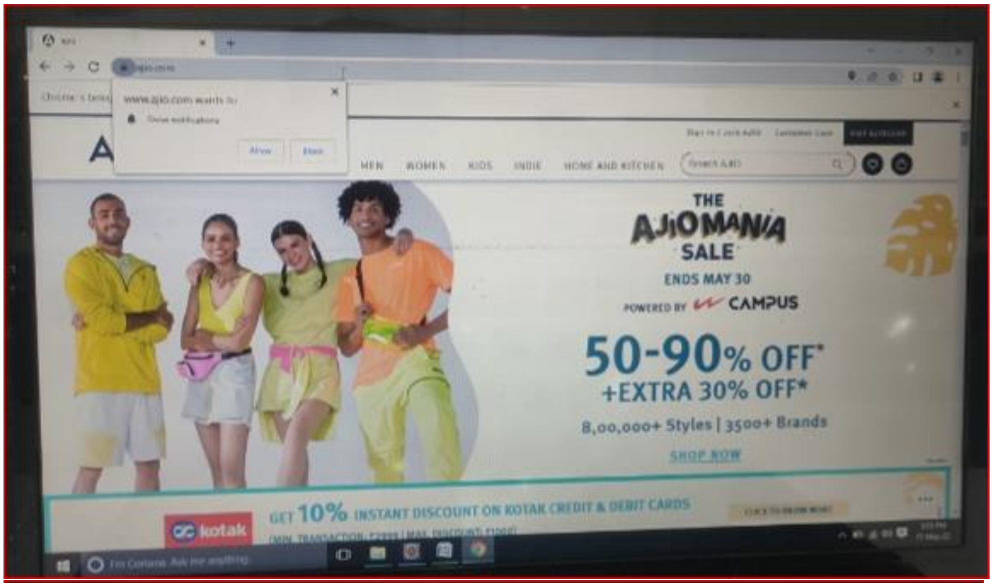
  }

}
```

OUTPUT :




Chrome is being controlled by automated test software.
AJIO
MEN
WOMEN
KIDS
INDIE
HOME AND KITCHEN
THE
AJIO MANIA
SALE
ENDS MAY 30
POWERED BY CAMPUS
50-90% OFF*
+EXTRA 30% OFF*
8,00,000+ Styles | 3500+ Brand
SHOP NOW
kotak
GET 10% INSTANT DISCOUNT ON KOTAK CREDIT & DEBIT CARDS


MEN WOMEN KIDS INDIE HOME AND KITCHEN
THE
AJIOMANIA
SALE
ENDS MAY 30
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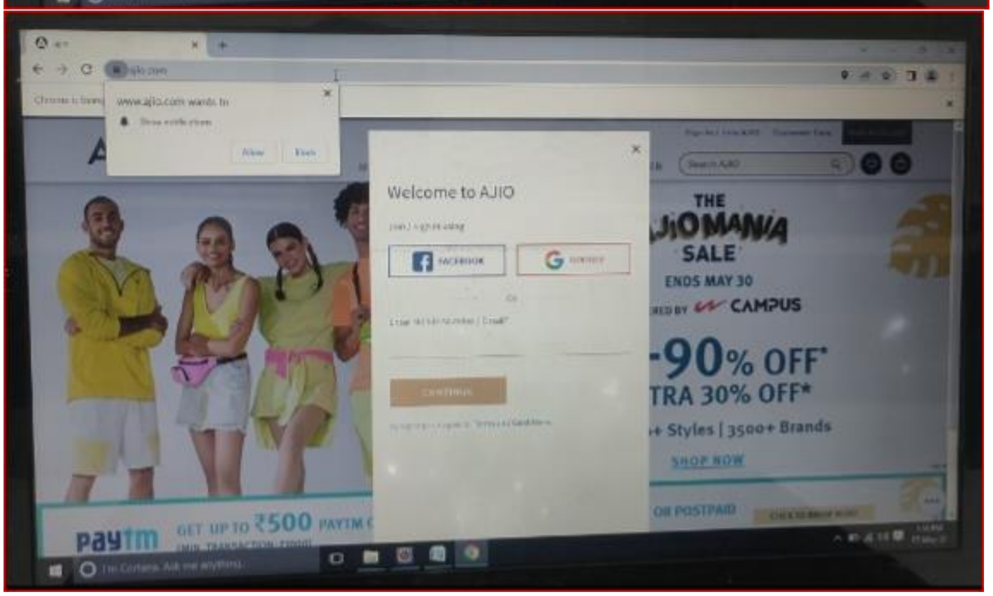
Welcome to AJIO
FACEBOOK
THE
SALE
ENDS MAY 30
CAMPUS
SHOP NOW
Paytm
GET UP TO ₹500
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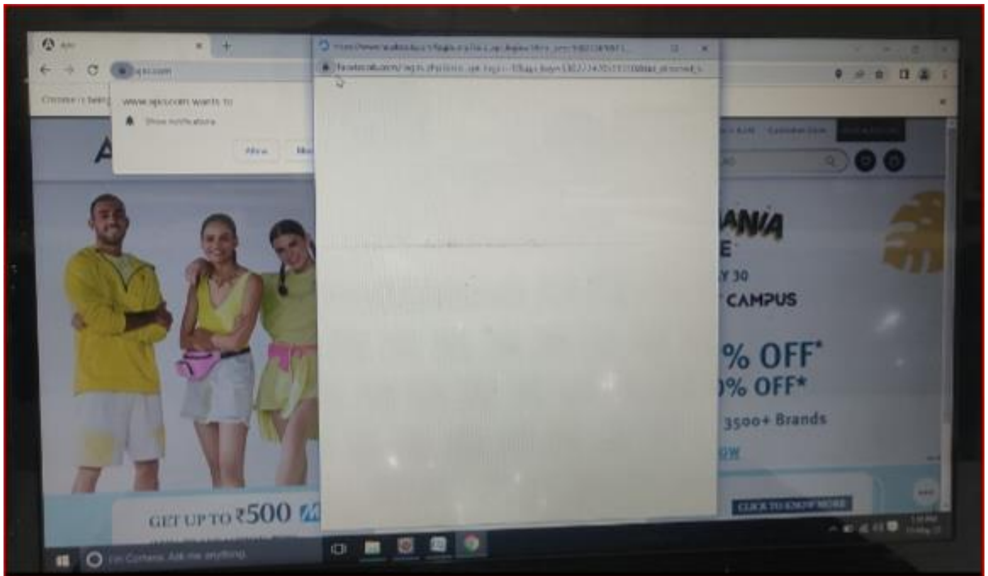
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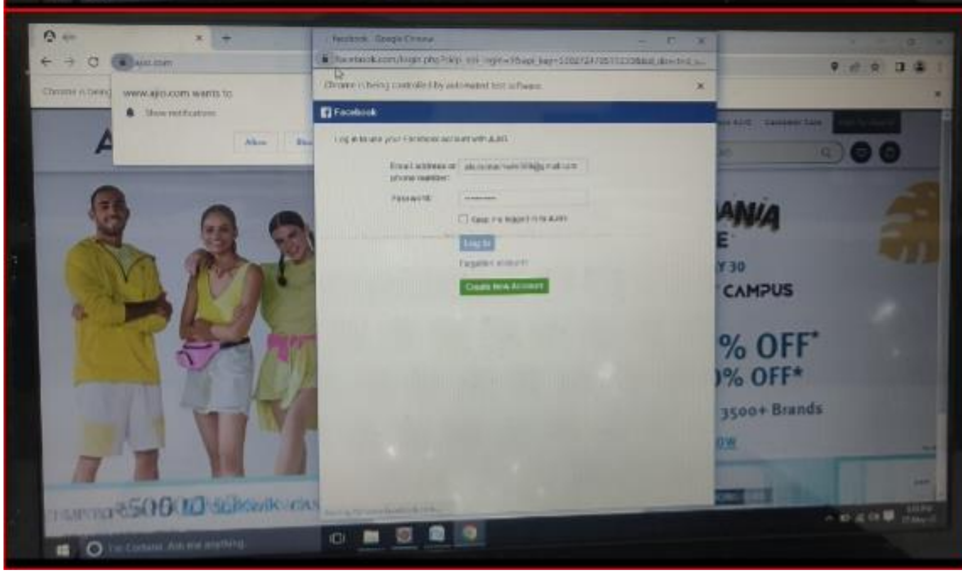
www.ajio.com wants to
Show notifications
Chrome is being controlled by automated test software.
Facebook
CAMPUS
3500+ Brands
Create New Account